ZS-F34
The Six-Dimensional Charge Unit of the Z-Spin Three-Form

The Icosahedral Bivector Selection of the Baryon Module and the Finite-Symmetry Susceptibility-Rank Theorem, with the Single-Mode Reduction of the Dark-Energy Susceptibility in the Exact-A_5 Branch

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Verification: 23/23 (zs_f34_verify_v1_1.py) + 32/32 gate-decomposition-v5 (zs_f34_verify_v1_8.py) PASS (arithmetic, counterexample, and model-instantiation checks — **not** a closure certificate). **This is the terminal version of ZS-F34.** Two general theorems are placed at the front, lifting the result from a conditional internal coefficient to a statement of independent interest. (A) **Icosahedral Bivector Selection (F34.BIV):** the exterior square of the standard four-dimensional A_5-module is $\Lambda^2(\mathbf{4}) = \mathbf{3} \oplus \mathbf{3'}$ (a textbook character identity, GC27); it is the six-dimensional *irreducible* of S_5 (GC28); the two triplets are the Hodge **self-dual** and **anti-self-dual** sectors, and an orientation-reversing odd permutation in S_5 commutes/anticommutes with the Hodge star so as to *swap* them (GC29). This **geometrizes** the outer automorphism σ of v1.6: $3 \leftrightarrow 3'$ *is* self-dual ↔ anti-self-dual under orientation reversal. Conditional on identifying the corpus rank-6 Y/baryon carrier with $\Lambda^2(V_4)$, the module $\mathbf{3} \oplus \mathbf{3'}$ is selected — the mathematics is DERIVED-CONDITIONAL, while the physical identity of the orientation-reversing symmetry remains OPEN. (B) **Finite-Symmetry Susceptibility-Rank Theorem (F34.SR):** for a finite group G acting unitarily on a flux space V, a G-equivariant kinetic operator $Z > 0$, and a source $c \in V^G$, the susceptibility is $\chi = (1/4\pi^2)\, c^\dagger (Z|_{V^G})^{-1} c$, so the *effective mode count is dim V^G, not the full flux rank* (GC30). The single-mode dark-energy result ($\dim V^G = 1$) and the m×m block case are its corollaries; the "factor 83" was never the right multiplicity for an invariant source. **A residual v1.6 error is fixed:** the single-mode susceptibility is $\chi_- = (Z_{match}\, g_{reg}^2/4\pi^2\, V_\Sigma)\, e_6^2\, (\nu_s^2/G_s)$ with $\nu_s^2 = v_s^T v_s$ (lattice norm) and $G_s = s^T G s$ (Hessian norm), since $v_s^T G^{-1} v_s = \nu_s^2/G_s$ already — v1.6 divided by G_s twice (GC23/GC32). The 83-dimensional flux/source realization reduces to **five one-dimensional objects** $\{P_s, G_s, \nu_s, (U_s, V_s) = 1, Q_s\}$; the independent G-Metric (V_Σ) and G-Charge (Z_{match}, e_6) gates remain, and the actual P_b, I_s, e_6 are deferred to ZS-F35/ZS-F36. **v1.8 (this version)** is an eighth-review final-clean pass: it simplifies math spacing, sharpens F34.SR to $N_{eff} = \dim S_{src} \le \dim V^G$ (equality when admissible sources span V^G), adds the Euclidean-vs-Lorentzian **signature caveat** to F34.BIV (its $\star^2 = +1$ real split is identified with the Lorentzian $\Lambda^2(\mathbb{R}^{1,3})$ only via complexification/an explicit intertwiner, part of the carrier bridge), and records that F34.BIV selects the *representation type* $3 \oplus 3'$ while the actual baryon projector P_b retains a residual $\mathbb{C}P^3$ **multiplicity-line freedom** (→ ZS-F35). No new theorems or closures are added. ($\mathbf{A}$, $\mathbf{Q}$, dim $\mathbf{Z}$) = (35/437, 11, 2) **LOCKED**.

§0. Abstract

This is the companion paper promised by ZS-F33 §9.5, revised after external review. ZS-F33 reduced the absolute cosmological vacuum scale to a single dimensionful susceptibility through $\rho_{\Lambda,Z} = \frac{1}{2} \chi_- \omega^2$, and registered the *Charge-Unit Obstruction*: flux integrality fixes the flux number but not the dimensionful unit $\chi_- = e_-^2/(4\pi^2 Z_-)$. F33.8D recorded a route to χ_- under a rank-2 internal-cycle ansatz $Y_6 = M_4 \times \Sigma_2$ and deferred its establishment to this paper.

What is established. The uplift is **internal-fibre, not Kaluza–Klein**: the "6" is the representation dimension $\dim \Lambda^2(\mathbb{R}^{1,3}) = 6$, and a 6D *spacetime* metric is **not licensed** (ZS-A17; asserting one triggers the F33-G8D retraction). A **degree-counting argument** (§2) shows the 4D odd three-form A_3^- (field strength F_4^-, a 4-form) descends from a parent five-form C_5 (six-form strength $G_6 = dC_5$) by wrapping the **unique** 2-cycle ($b_2 = 1$); a 6-form needs a carrier of dimension **≥ 6**, and $M_4 \times \Sigma_2$ (dim 4 + dim **Z** = 6 = **Y**) *saturates* this, so it is the **minimal** carrier — the product of the *existing* 4D base and the *existing* Z-sector Koenigs torus $\Sigma_2 = E_\lambda^*$, a six-dimensional pseudo-Riemannian **total space** (four noncompact spacetime dimensions + a compact 2D internal fibre), **not** six observable spacetime dimensions. A **parent-action wrapped-brane reduction** of $C_5 = A_3^- \wedge \omega_2$ gives the 4D kinetic matrix $Z_{ij} = (1/g_6^2)\int_{\Sigma_2} \omega_2^i \wedge \star_2\omega_2^j$ and the susceptibility in its **general quadratic form** in the theta-coupling vector $\mathbf{c}_\theta$

$$\chi_- = \frac{1}{4\pi^2} \mathbf{c}_\theta^{\mathsf T} Z^{-1}\mathbf{c}_\theta.$$

What v1.7 adds (terminal version). v1.6 corrected an A_5 trichotomy and proposed an outer-automorphism selection; a seventh review found one real formula error (fixed below) and recommended terminating F34. A deep exploration then showed that the result's value is raised most not by closing another gate but by stating two general theorems. v1.7 does both.

- **Icosahedral Bivector Selection (F34.BIV, new — geometrizes v1.6's σ).** The standard four-dimensional A_5-module V_4 (the sum-zero subspace of the S_5 permutation representation) has exterior square $\Lambda^2(V_4) = \mathbf{3} \oplus \mathbf{3'}$ — an elementary character computation ($\chi_{\Lambda^2 4} = (6,-2,0,1,1) = \chi_3 + \chi_{3'}$, GC27). Over S_5 this is the six-dimensional *irreducible* (GC28), so the two A_5-triplets cannot be S_5-separated: an odd permutation mixes them. Geometrically, in an oriented 4-dimensional metric space $\Lambda^2(V_4) = \Lambda^2_+ \oplus \Lambda^2_-$ (self-dual ⊕ anti-self-dual, each 3-dimensional); $A_5 \subset SO(V_4)$ commutes with the Hodge star and preserves each, while an orientation-reversing odd permutation anticommutes with it and **swaps $\Lambda^2_+ \leftrightarrow \Lambda^2_-$** (GC29). This is exactly v1.6's outer automorphism, now *realized* rather than posited: **$\mathbf{3} \leftrightarrow \mathbf{3'}$ is self-dual ↔ anti-self-dual under orientation reversal**. The same $3 \oplus 3'$ is the known decomposition of the six-dimensional crystallographic icosahedral representation ($T_1 \oplus T_2$, IMPORTED-PROVEN), an independent cross-check. Conditional on identifying the corpus rank-6 Y/baryon carrier with $\Lambda^2(V_4)$, the module $\mathbf{3} \oplus \mathbf{3'}$ is selected with no extra representation choice. The mathematics is DERIVED-CONDITIONAL; the physical identity of the orientation-reversing symmetry stays

OPEN (split from v1.6's single HYPOTHESIS-strong tag into F34.OUT-Math and G-Outer-Physical).

- **Finite-Symmetry Susceptibility-Rank Theorem (F34.SR, new — the external-value statement).** Let a finite group G act unitarily on a flux space V, let $Z > 0$ be G-equivariant, and let the source $c \in V^G$. Then Z^{-1} is G-equivariant, it preserves $V = V^G \oplus (V^G)^{\perp}$, and since $c \in V^G$,

$$\chi = \frac{1}{4\pi^2}\mathbf c^{\dagger}\big(Z|_{V^G}\big)^{-1}\mathbf c,\qquad N_{\rm eff} = \dim V^G\ \ (\neq \dim V).$$

The *effective mode count is the invariant-source rank, not the full flux rank* (GC30). Three corollaries: $\dim V^G = 0 \Longrightarrow c = 0$; $\dim V^G = 1 \Longrightarrow$ exact single mode; $\dim V^G = m > 1 \Longrightarrow$ an $m\times m$ invariant-block susceptibility. This is a statement about any finite-symmetry multi-four-form theory (Bousso–Polchinski-type included): the naive "N fluxes $\Longrightarrow$ susceptibility $\propto$ N" is generally false. In F34 the baryon/CDM-equivariant single-singlet branch has $\dim H_\Lambda^{A_5} = 1$, so $N_{\rm eff} = 1$ and full 83-dimensional centrality is *not needed* — the old question "why are 83 kinetic eigenvalues equal?" is replaced by "what is the one-dimensional invariant-source response?".

One real error fixed. v1.6 wrote $\chi_- = (Z_{\rm match}\, g_{\rm reg}^2/4\pi^2 V_\Sigma)\, e_6^2\, (\nu_s^2/G_s)$ **and** defined $\nu_s^2 = v_s^{\mathsf T}G^{-1}v_s$, which double-divides by G_s ($v_s^{\mathsf T}G^{-1}v_s$ already equals ν_s^2/G_s). v1.7 separates the **lattice norm** $\nu_s^2 = v_s^{\mathsf T}v_s = \langle v_s, v_s\rangle_0$ from the **Hessian norm** $G_s = s^{\mathsf T}G\, s$ (s the unit singlet), so that $q_s := v_s^{\mathsf T}G^{-1}v_s = \nu_s^2/G_s$ and

$$\chi_-^{(s)} = \frac{Z_{\rm match}g_{\rm reg}^2}{4\pi^2V_\Sigma}e_6^2\frac{\nu_s^2}{G_s} = \frac{g_{\rm reg}^2e_6^2}{4\pi^2V_\Sigma}I_s,\qquad I_s := Z_{\rm match}v_s^{\mathsf T} G^{-1} v_s,$$

introducing the **Singlet Response Invariant** I_s, one basis-independent number bundling $Z_{\rm match}$, the singlet Hessian, and the primitive-source normalization (GC23, GC32, robust over many A_5-invariant metrics).

Net. The robust object is the master quadratic form $\chi_- = (Z_{\rm match}\, g_{\rm reg}^2/4\pi^2 V_\Sigma)\, e_6^2\, \hat c_\theta^{\mathsf T}G^{-1}\hat c_\theta$; F34.SR fixes its effective rank at $\dim V^G$, F34.BIV selects the rank-6 carrier, and the exact-A_5 branch reduces to I_s plus the independent G-Metric (V_Σ) and G-Charge ($Z_{\rm match}$, e_6) gates. The number $0.091847 \cdot e_6^2$ ($\rho_{\Lambda,Z} = 0.234402 \cdot e_6^2$) is the path-(c) aligned-central value only; single-mode is **one of three branches**, not generic. **The actual P_b, I_s, and e_6 are deferred to ZS-F35; F34 terminates here.** The contribution is a *reduction*, not a closure. (**A**, **Q**, dim **Z**) = (35/437, 11, 2) **LOCKED**.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN / IMPORTED-PROVEN	Complete proof, here or in cited peer-reviewed work.
DERIVED	Follows from Z-Spin structure + standard mathematics; no fitted parameter.
DERIVED-CONDITIONAL	Derived conditional on a stated, not-yet-closed hypothesis.
HYPOTHESIS-strong	Multiple independent structural anchors and a documented promotion path, but a load-bearing identification still OPEN.
COMPUTED-UNDER-NORMALIZATION	A definite number, but only after an undetermined normalization is fixed by hand.
BENCHMARK-COROLLARY	A value that holds under an explicit, declared set of normalizations.
NO-GO / CLOSED-NEGATIVE	A registered impossibility, or a route resolved in the negative (scope stated).
METRIC-IDENTIFICATION-CONDITION	A definite value, conditional on identifying the physical metric with a canonical candidate.
MINIMALITY-CONDITIONAL	A structure derived as the minimal one of its kind, unique only under a no-extra-structure principle.
REDUCED	A gate shown to reduce to an upstream gate, inheriting that gate's (still-open) status rather than closing independently.
OPEN-GATE / SUB-GATE	A named, load-bearing gap the paper deliberately does **not** close (a sub-gate is one component of a decomposed gate).
RETRACTED	A claim withdrawn in a later version, with the reason recorded.
LOCKED	Core constant fixed upstream; immutable downstream.

Claim Ledger (17 entries: 6 results incl. two new general theorems, the A_5-path theorems, the F34.OUT split into Math + Physical, the metric/flux theorems one retracted, and the torsion-No-Go retraction)

Tag	Result	Status
F34.BIV	**Icosahedral Bivector Selection:** $\Lambda^2(V_4) = \mathbf{3} \oplus \mathbf{3'}$ for the standard 4-dim A_5-module V_4 (character, GC27); it is the 6-dim irreducible of S_5 (GC28); the triplets are Hodge self-dual/anti-self-dual, swapped by an orientation-reversing odd permutation (GC29). Conditional on identifying the corpus rank-6 Y/baryon carrier with $\Lambda^2(V_4)$, the *representation type* $3 \oplus 3'$ is selected (the actual projector P_b keeps a residual $\mathbb{C}P^3$ multiplicity-line freedom → F35); Euclidean $\star^2$=+1 vs Lorentzian carrier via complexification (signature caveat)	PROVEN (math) / DERIVED-CONDITIONAL (carrier)
F34.SR	**Finite-Symmetry Susceptibility-Rank Theorem:** for G-equivariant Z > 0 and source c ∈ V^G, $\chi = (1/4\pi^2)\, c^\dagger(Z\|_{V^G})^{-1} c$, so N_eff = dim V^G ≠ dim V (GC30). Corollaries: dim V^G = 0 ⟹ c = 0; = 1 ⟹ single mode; = m ⟹ m×m block	PROVEN (general)
F34.1	Internal-fibre $Y_6 = M_4 \times \Sigma_2$: a 6-form needs dim ≥ 6, so $M_4 \times \Sigma_2$ (4 + dim **Z** = 6) is the **minimal** carrier (6-dim total space, not a 6D spacetime)	DERIVED (minimal) / MINIMALITY-CONDITIONAL (unique)
F34.2	Parent-action wrapped-brane reduction ⟹ Z_{ij}, and $\chi_- = (1/4\pi^2)\, \hat{\mathbf{c}}_\theta^T Z^{-1} \hat{\mathbf{c}}_\theta \cdot e_6^2$ (theta-coupling, distinct from Q_source)	DERIVED-CONDITIONAL (anomaly gates)
F34.M1	λ*-*invariance alone* fixes τ but **not** the Kähler area (c_Σ free)	NO-GO (PROVEN)
F34.M2	v1.2 "Koenigs coordinate forces c_Σ = 1"	**RETRACTED** (non sequitur)
F34.M3	**Principal-polarization candidate metric:** Θ_Z ($c_1 = 1$) ⟹ representative Area = 1, V_Σ = 1	METRIC-IDENTIFICATION-CONDITION
F34.4	$g_{reg}^2 = 6\mathbf{A}/\mathbf{Q} = 210/4807$ (ZS-M6); EFT match Z_match (in the **denominator** of Z, numerator of χ_-)	DERIVED-CONDITIONAL (matching)

Tag	Result	Status
F34.4-NG	v1.0 "even-dim Ray–Singer torsion" No-Go	**RETRACTED** (category error)
F34.M4	**G-Flux decomposition:** diagonality **DERIVED-COND**; **centrality OPEN** (needed only in path c); A_5 commutant $M_4 \oplus M_4 \oplus M_6 \oplus M_7$ (**dim 117**); 121 = integral-lattice embedding (full index OPEN)	DECOMPOSED (centrality OPEN)
F34.A5-1	$\mathrm{End}(V_{11}) = 3\cdot 1 \oplus 6\cdot 3 \oplus 6\cdot 3' \oplus 8\cdot 4 \oplus 10\cdot 5$, {baryon+DE} = End−CDM = $1\cdot 1 \oplus 4\cdot 3 \oplus 4\cdot 3' \oplus 6\cdot 4 \oplus 8\cdot 5$ (one singlet)	DERIVED (chars) / CONDITIONAL (baryon OPEN)
F34.SEL	**Singlet-selection (conditional on P_b, P_c orthogonal A_5-equivariant projectors):** a nonzero A_5-invariant θ-coupling exists **iff baryon ≠ $1 \oplus 5$**; singlet-free candidates $\{2\cdot 3, 3 \oplus 3', 2\cdot 3'\}$ — A_5 alone does **not** uniquely pick $3 \oplus 3'$	DERIVED-CONDITIONAL (excludes $1 \oplus 5$)
F34.OUT-Math	$\sigma \in \mathrm{Out}(A_5)$ swaps $3 \leftrightarrow 3'$, fixes 1,4,5; only $\mathbf{3 \oplus 3'}$ is σ-stable among singlet-free candidates ⇒ A_5 + σ-stability + no-baryon-singlet **uniquely** select $3 \oplus 3'$ (realized geometrically by F34.BIV)	DERIVED-CONDITIONAL
G-Outer-Physical	Identifying σ with a physical Z-Spin symmetry (seam orientation reversal / J_Z parity / register automorphism / CPT exchange)	**OPEN**
F34.A5-2	**Single-mode reduction (path a):** if DE keeps the singlet **and** $[K, R(g)] = 0$, $\chi_- = (Z_{match}\, g_{reg}^2/4\pi^2\, V_\Sigma)\, e_6^2\, (\nu_s^2/G_s)$ with $\nu_s^2 = v_s^T v_s$, $G_s = s^T G\, s$, so $q_s = v_s^T G^{-1} v_s = \nu_s^2/G_s$; **factor 83 replaced**. Bundled as $I_s := Z_{match}\, v_s^T G^{-1} v_s$. If A_5 broken, Schur complement	DERIVED-CONDITIONAL (exact-A_5 + kinetic A_5); I_s value OPEN
F34.6	Master $\chi_- = (Z_{match}\, g_{reg}^2/4\pi^2\, V_\Sigma)\, e_6^2\, \hat{\mathbf{c}}_\theta^T G^{-1} \hat{\mathbf{c}}_\theta$; value $0.091847\cdot e_6^2$ only in path (c) at $\{V_\Sigma=1, G=I, \hat{\mathbf{c}}_\theta=\text{uniform}, Z_{match}=1\}$	CANDIDATE (path c, aligned)
F34.8	Robust: the master quadratic form with $N_{eff} = \dim V^G$ (F34.SR); exact-A_5 branch reduces to I_s + independent G-Metric (V_Σ) and G-Charge (Z_{match}, e_6); dimensionful residual e_6	OPEN (terminus; P_b, I_s, e_6 → ZS-F35)

The thermal-history material (formerly F34.7) is moved to the Outlook (§10), as it does not feed the charge-unit reduction.

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§1. Introduction

ZS-F32 proved that the absolute vacuum scale reduces to a single odd-sector susceptibility through $\rho_{\Lambda,Z} = \frac{1}{2} \chi_- \omega^2$, $\omega = \arg \lambda^* = 2.2592495540$. ZS-F33 then showed *why* (**A**, **Q**) plus topology cannot finish: the energy spacing $E_k(\theta) = (e_-^2/2Z_-)(k + \theta/2\pi)^2$ depends on the **dimensionful unit** $\chi_- = e_-^2/(4\pi^2 Z_-)$, which flux integrality does not constrain (the *Charge-Unit Obstruction*, F33.8). F33.8D sketched the route — $Y_6 = M_4 \times \Sigma_2$, $C_5 = A_3^- \wedge \omega_2$ — and named ZS-F34 as the paper that would establish it.

The first release (v1.0) claimed that three of the four unknowns $\{V_\Sigma, g_6, N, e_6\}$ close, leaving e_6 as the sole residual. Seven successive external reviews progressively dismantled the over-claims. v1.1 reframed the result as three named gates. v1.2 claimed to close two; **v1.3 retracted both** (the Koenigs argument is a non sequitur; the flux coefficient hid a 121-fold ambiguity). **v1.4** corrected the centrality and carrier errors and introduced an A_5 analysis. **v1.5** corrected that analysis (Schur is four *multiplicity matrices*, commutant 117, not four scalars; the 83-decomposition is baryon-conditional; Z_match was inverted) and computed a source trichotomy instead of a No-Go. **v1.6** corrected three v1.5 over-reaches (singlet-selection only *excludes* baryon = $1 \oplus 5$; the single-mode reduction needs the *kinetic* operator A_5-invariant; the singlet *normalization* ν_s is not fixed by A_5) and proposed an outer-automorphism selection plus a single-mode reduction. **v1.7 (this terminal version) fixes one real formula error and places two general theorems at the front:** the **Icosahedral Bivector Selection Theorem** (F34.BIV: $\Lambda^2 V_4 = 3 \oplus 3'$, which *geometrizes* v1.6's outer automorphism as the Hodge self-dual ↔ anti-self-dual swap under orientation reversal) and the **Finite-Symmetry Susceptibility-Rank Theorem** (F34.SR: for a G-equivariant kinetic operator and a source in V^G, the effective mode count is $\dim V^G$, not the full flux rank — a statement of independent mathematical-physics interest). The one error fixed is the ν_s^2/G_s double-counting in the single-mode susceptibility. The robust deliverable is the master quadratic form, with every unresolved gate located in an explicit factor; the residual computations (P_b, the Singlet Response Invariant I_s, and e_6) are deferred to ZS-F35.

Dependencies: ZS-F33 (obstruction, F33.8D route, the Koenigs-torus complex structure and theta-line $c_1 = 1$), ZS-F32 ($\frac{1}{2}\chi_- \omega^2$), ZS-M1 ($z^*$, λ^*), ZS-F5/F18 ((Z,X,Y) = (2,3,6)), ZS-M6 ($\kappa_{\rm reg}^2 = \mathbf{A}/\mathbf{Q}$, Dimensional Coupling Norm Theorem), ZS-M11/S1 (the A_5 = icosahedral-rotation representation theory of the truncated-icosahedron Hodge complex), ZS-S4 (C_M^{sp}, kept distinct), ZS-A17 (Spin–Metric No-Go), ZS-A24 (the type-II_1 corner and the continuous-trace face projectors), ZS-A28 (the rank/energy separation and the lattice no-go), ZS-A29/A30 (the rank-83 kinematic precursor and one-quantum-per-face). External: Bousso–Polchinski, Brown–Teitelboim, Ray–Singer, Milnor, Quillen/Mumford, Appell–Humbert.

Notation (corrected in v1.1). Two distinct constants are never written with the same symbol again:

$$\kappa_\lambda := -\ln|\lambda^*| = 0.1148346250\quad(\text{loxodromic decrement}),\qquad \kappa_{\rm reg}^2 := \frac{\mathbf{A}}{\mathbf{Q}} = \frac{35}{4807}\quad(\text{register normalization}).$$

These differ numerically ($6\kappa_\lambda^2 = 0.0791 \neq 6\mathbf{A}/\mathbf{Q} = 0.0437$), and v1.0's shared symbol "κ" is a genuine source of confusion now removed. The gauge coupling is $g_{reg}^2 = 6\kappa_{reg}^2$, and the torus area uses κ_λ.

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§2. The Internal-Fibre Uplift (F34.1)

2.1 The "6" is a representation dimension

By F33.3A′, equating the complex-doubling count $2(d-1)$ with the bivector count $d(d-1)/2$ gives $d = 4$, $X = 3$, $\mathbf{Y = 6 = \dim \Lambda^2(\mathbb{R}^{1,3}) = \dim \mathfrak{so}(1,3)}$ [**AC3**, PROVEN]. This is the dimension of the Y-sector *field representation*, not a count of spacetime directions. (This distinction is used again, decisively, in §5.2.)

2.2 Why Kaluza–Klein is not licensed, and the internal-fibre route is forced

A Kaluza–Klein uplift would require an *independent* 6D spacetime metric. The present corpus does not license one: ZS-A17 Theorem F proves the spatial metric is **not reconstructible** from **A** or the spin structure, and ZS-F31 carries the spacetime metric as an imported datum. The corpus is explicit that asserting a 6D *spacetime* from the representation "6" is a **retraction trigger** (F33-G8D). We therefore read the "6" as the **internal-fibre product**

$$Y_6 = M_4 \times \Sigma_2,\qquad M_4 = \text{(existing 4D base)},\quad \Sigma_2 = \text{(existing Z-sector 2-cycle)} ,$$

a product of structures the corpus *already* has — never a new 6D spacetime.

Two facts make Σ_2 forced. **(i)** The bivector 2-form must integrate over a 2-cycle; the *only* 2-cycle is the Z-sector Koenigs torus E_λ^* ($\dim \mathbf{Z} = 2$, $b_2 = 1$; F33.2B PROVEN), so $\Sigma_2 \equiv E_\lambda^*$. **(ii)** Consistency with F32: F32 places $F_4^- = dA_3^-$ in 4D, and wrapping $C_5 = A_3^- \wedge \omega_2$ on Σ_2 reproduces exactly that A_3^- (§3).

2.3 The degree-counting derivation: $M_4 \times \Sigma_2$ as the *minimal* carrier (corrected in v1.4)

The factorization is strongly constrained by form degree, though — as a fourth review noted — it yields a *minimal* carrier, not a uniquely forced one. Start from what cosmology fixes: the odd sector is the 4D three-form A_3^- with field strength $F_4^- = dA_3^-$, a **4-form** (Kaloper–Sorbo; F32). Demand that A_3^- descend from a single parent gauge form by wrapping an internal harmonic form ω_k on the internal cycle, $G_{parent} = F_4^- \wedge \omega_k$. Then:

- the internal cycle is the unique 2-cycle Σ_2 ($b_2 = 1$), whose unique harmonic generator is a **2-form**, so $k = 2$;
- the parent field strength has degree $4 + k = \mathbf{6}$ (a 6-form $G_6 = dC_5$), so the parent potential C_5 is a **5-form**;
- a 6-form needs a carrier of dimension $\geq \mathbf{6}$. The product of the 4D base with the 2-cycle, $\dim(M_4 \times \Sigma_2) = 4 + \dim \mathbf{Z} = 6$, **saturates** this bound, so $M_4 \times \Sigma_2$ is the **minimal** such carrier.

What the degree count proves is therefore *minimality*, not uniqueness: a carrier of dimension > 6 with unused directions is not excluded by form degree alone. Uniqueness requires an added principle — **Minimal-Carrier** (no unused extra dimensions) or **Corpus-Completeness** (Σ_2 is the only internal cycle the corpus supplies). Under either, $M_4 \times \Sigma_2$ is selected [**GC15**].

Two cautions, both from review. First, the kinetic term $\int_{M_4\times\Sigma_2} G_6 \wedge \star_6 G_6$ genuinely uses a six-dimensional product metric and Hodge star, so the precise statement is: **$M_4 \times \Sigma_2$ is a six-dimensional pseudo-Riemannian total space with four noncompact spacetime dimensions and a compact two-dimensional internal fibre** — *not* a claim of six observable spacetime dimensions (F33-G8D). Second, the field-representation count $\dim \Lambda^2\mathbb{R}^{1,3} = 6$ and the carrier dimension $4 + \dim \mathbf{Z} = 6$ coincide, as do the corpus's bivector route $C(4,2) = 6$ and sector-product route $\dim \mathbf{X} \times \dim \mathbf{Z} = 3 \times 2 = 6$; but these are *distinct arithmetic operations*, and their agreement is an internal-consistency indicator, **not** a proof of "not numerology." The four routes reconfirm the Y representation dimension; they do not by themselves single out $M_4 \times \Sigma_2$ as the unique physical carrier.

> **Theorem F34.1 [DERIVED minimal / MINIMALITY-CONDITIONAL unique].** Producing the 4D odd three-form A_3^- by wrapping the unique internal 2-cycle forces a five-form C_5 whose 6-form strength needs a carrier of $\dim \geq 6$; $M_4 \times \Sigma_2$ ($4 + \dim \mathbf{Z} = 6 = \mathbf{Y}$) is the minimal carrier, and is *the* carrier under a no-extra-dimension / corpus-completeness principle. It is a 6-dim total space, not a 6D spacetime. **Gate G-A1:** identifying Σ_2 with a non-corpus 2-cycle, asserting a 6D spacetime metric, or claiming uniqueness without the minimality principle, voids this (F33-G8D).

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§3. The Parent-Action Wrapped-Brane Reduction (F34.2)

v1.0 began directly from a 4D membrane term and the single normalization $Z_- = V_\Sigma/g_6^2$. Review correctly noted that this skips the 6D→4D reduction and the anomaly bookkeeping that F33.8D required. v1.1 supplies the parent action and the reduction, and exposes the general (matrix) structure.

3.1 The parent action

We start in 6D with a five-form sector and wrapped 4-branes (5D worldvolume $W_3 \times \Sigma_2$):

$$S_6 = -\tfrac{1}{2}\int_{M_4\times\Sigma_2} Z_{IJ} G_6^I\wedge\star_6 G_6^J + \sum_a e_a^{(6)}\int_{W_3^{(a)}\times\Sigma_2} C_5^a ,\qquad G_6^I = dC_5^I .$$

3.2 Harmonic expansion and the reduced data

Expanding each C_5^I on a basis $\{\omega_2^i\}$ of $H^2(\Sigma_2)$ — for the geometric torus $b_2 = 1$, so a *single* ω_2 (the multiplicity question is deferred to §5) — with the theta-line cohomology normalization $\int_{\Sigma_2} \omega_2 = 1$ ($c_1 = 1$, F33.2B Computation Θ):

$$C_5^I = \sum_i A_3^{(i)}(x)\wedge\omega_2^i(y),\qquad G_6^I = \sum_i F_4^{(i)}(x)\wedge\omega_2^i(y),$$

the kinetic term and the wrapped-brane term reduce, factorizing base and fibre, to

$$Z_{ij}^{(4)} = \int_{\Sigma_2} Z_{IJ} \omega_2^i\wedge\star_2\omega_2^j = \frac{1}{g_6^2}\int_{\Sigma_2}\omega_2^i\wedge\star_2\omega_2^j,\qquad e_i^{(4)} = e_i^{(6)}\int_{\Sigma_2}\omega_2^i = e_i^{(6)},\qquad T_{\rm mem}^{(4)} = T_{\rm 4\text{-}brane}^{(6)}\cdot\mathrm{Area}(\Sigma_2) .$$

So the 4D membrane charge **$e_{mem} = e_6^{(6)}$** (with the unit-normalized ω_2), and its tension is the 6D 4-brane tension times the cycle area. The compact-3-form energy of F33 follows, with the kinetic normalization now a **matrix** Z_{ij}, giving

$$\chi_- = \frac{1}{4\pi^2} \mathbf{c}_\theta^{\mathsf T} Z^{-1}\mathbf{c}_\theta$$

which reduces to the F33.8D scalar $\chi_- = e_{mem}^2/(4\pi^2 Z)$ only in the one-flux ($b_2 = 1$) case.

3.3 Anomaly gates

F33.8D required the wrapped sector to be anomaly-free. The reduction is therefore conditional on:

> **Theorem F34.2 [DERIVED-CONDITIONAL on the anomaly gates].** $\chi_- = (1/4\pi^2)\, \mathbf{c}_\theta^T Z^{-1}\, \mathbf{c}_\theta$, with ($Z_{ij}$, e_{mem}, T_{mem}) as above. **Gate G-A2 (anomaly):** the charge lattice is unimodular; the membrane tadpole cancels; the global (mod-2) anomaly vanishes; orientation reversal acts as the J_Z-odd parity; membrane nucleation changes the flux by exactly one unit. Any failure voids the reduction. *None of these is verified here* — they are the executable content of F33.8D's "anomaly-free wrapped sector."

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§4. The Metric Gate: No-Go (F34.M1), the Retracted Koenigs Argument (F34.M2), and the Principal-Polarization Metric (F34.M3, F34.3)

4.1 F34.M1: λ*-invariance alone does not fix the Kähler area

> **Theorem F34.M1 (Metric-Normalization Obstruction) [NO-GO, PROVEN].** *λ*-invariance alone* fixes the *complex structure* of E_{λ^*} ($\tau = (\omega + i\kappa_\lambda)/2\pi$, $\mathrm{Im}\,\tau > 0$; F33.2B, DERIVED) but **not** the Kähler *area*. *Proof.* For any constant $c_\Sigma > 0$ the metric $g_{c_\Sigma} = c_\Sigma^2|d\log w|^2$ is invariant under $w \mapsto \lambda^* w$ (a constant rescaling commutes with the discrete action), with Area $= c_\Sigma^2 \cdot 2\pi\, \kappa_\lambda$ [**AC11**]. So invariance leaves a one-parameter family.

The corpus theta-line normalization **$c_1 = 1$** ($\int_{\Sigma_2} \omega_2 = 1$; F33.2B Computation Θ) is metric-*independent* and does not fix the area either [**AC10**] — *as a number*. Its role is deeper than v1.1/v1.2 used: it fixes the **degree** of the polarization (§4.3).

4.2 Retraction of the v1.2 Koenigs argument (F34.M2)

v1.2 argued (theorem F34.M2) that the Koenigs linearizing coordinate forces $c_\Sigma = 1$, because $c_\Sigma^2|d\log w|^2$ supposedly corresponds to the power coordinate w^{c_Σ} (whose multiplier $\lambda^{*c_\Sigma} \neq \lambda^*$ fails to linearize). External review correctly identified this as a **non sequitur**:

> **F34.M2 [RETRACTED].** Choosing the Riemannian metric $g_{c_\Sigma} = c_\Sigma^2|d\log w|^2$ on the *same* Koenigs coordinate w is **not** the same operation as changing coordinates to w^{c_Σ}. For any $c_\Sigma > 0$ the metric g_{c_Σ} is defined on the unchanged coordinate and remains invariant under $w \mapsto \lambda^* w$. Koenigs' theorem fixes the linearizing coordinate up to a nonzero constant — the *conformal* structure — but the overall *Kähler scale* of the metric placed on that coordinate is a separate datum it does not determine. This confirms only that $\lambda^{*c_\Sigma} \neq \lambda^*$, which is irrelevant to whether $c_\Sigma^2|d\log w|^2$ is admissible. The Koenigs route is therefore withdrawn.

So after F34.M1 alone, V_Σ reverts to **COMPUTED-UNDER-NORMALIZATION** — pending an *independent* metric-selection principle.

4.3 F34.M3: the principal polarization of the theta line fixes the scale (G-Metric)

That principle is the **polarization** the corpus already carries. F33.2B established the degree-1 theta line Θ_Z with **$c_1(\Theta_Z) = 1$** ("fibre flux," $\int_Z \mathrm{curv} = 1$), kept strictly separate from the flat Wilson line ($c_1 = 0$). By the **Appell–Humbert theorem**, line bundles on the complex torus $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ are classified by a Hermitian form H with $\operatorname{Im} H(\Lambda, \Lambda) \subseteq \mathbb{Z}$; a **degree-1 positive** such bundle is a **principal polarization**, with

$$H(z,w) = \frac{z \bar w}{\operatorname{Im}\tau},\qquad |c_1| = |\operatorname{Im}H(1,\tau)| = 1.$$

Its Kähler metric and form are $ds^2_\Theta = |dz|^2/\operatorname{Im}\tau$ and $\omega_\Theta = (1/\operatorname{Im}\tau)\, dx \wedge dy$, and the key point is that the **area equals the degree**:

$$\operatorname{Area}(E_\tau) = \int_{E_\tau}\omega_\Theta = c_1 = 1.$$

This is a **topological** equality — the area is the first Chern number, not a free Kähler modulus — which is exactly the datum the Koenigs argument lacked. (In the $u = \log w$ coordinate the same metric reads $ds^2_\Theta = |du|^2/(2\pi\,\kappa_\lambda)$; its conformal factor $1/(2\pi\,\kappa_\lambda) = 1.385949$ is *numerically* the v1.2 "V_Σ", but the invariant area is 1.) For the unit-class 2-form ω_2 the Hodge norm on this representative is therefore

$$V_\Sigma = \int_{\Sigma_2}\omega_2\wedge\star_2\omega_2 = \frac{1}{\operatorname{Area}} = 1.$$

Convention. We use ω_Θ with $\int\omega\_\Theta = c_1$ in the normalization $c_1 = [\mathrm{Im}\ H]$ (equivalently $c_1 = [i\ F/2\pi]$ for the Chern connection F); the corpus theta-line statement $\int\_Z$ curv = 1 (F33.2B) is in the same normalization, so no 2π ambiguity enters between "fibre flux = 1" and "Area = 1."

v1.4 separates **three** distinct claims that v1.3 ran together:

1. the principal-polarization **class** is integral of degree 1 — **IMPORTED-PROVEN** (Appell–Humbert + F33.2B's $c_1 = 1$);
2. the chosen translation-invariant **representative** has normalized area 1, so $V\_\Sigma = 1$ — **DERIVED** (a computation on that representative);
3. that this representative is the **physical kinetic Hodge metric** — a **METRIC-IDENTIFICATION-CONDITION**, not a theorem.

> **Theorem F34.M3 (Principal-Polarization Candidate Metric) [METRIC-IDENTIFICATION-CONDITION].** On the polarization representative of the degree-1 theta line Θ_Z, Area = c_1 = 1 and V_Σ = 1. The polarization class (degree 1) is imported-proven and its representative area is derived; identifying it with the physical Hodge metric is the residual postulate. This is a **candidate normalization** of the metric scale, not a closure.

Anti-numerology. $V\_\Sigma = 1$ is the statement Area = c_1 = 1 on the representative; the v1.2 near-miss to 2 ln 2 (which lived in the *conformal factor* 1.385949) is irrelevant to V_Σ and is **dropped**.

> **F34.3 [DERIVED on the representative / identification-conditional].** $V\_\Sigma = 1$ on the principal-polarization representative (F34.M3). **Gate G-Metric:** reduced to the metric-identification condition — the polarization representative is the canonical candidate, but its identification with the physical metric is a postulate. The coefficient consequence (§7) is the *only* metric-dependence of χ_-.

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§5. The Coupling and the Retraction of the Torsion No-Go (F34.4, F34.4-NG)

5.1 The register coupling, and the missing matching

ZS-M6's **Dimensional Coupling Norm Theorem** (Peter–Weyl + rank-1 β_0, PROVEN) gives the *finite-register* block-Laplacian coupling

$$g_{\rm reg}^2 = \dim(Y)\cdot\kappa_{\rm reg}^2 = 6\cdot\frac{\mathbf{A}}{\mathbf{Q}} = \frac{210}{4807} = 0.04368629\qquad[\textbf{AC16}].$$

But — as review noted — that the *register* coupling equals the *continuum five-form* kinetic coefficient in S_6 is a **separate matching proposition**. It requires a quadratic-response identity

$$\left.\frac{\delta^2\Gamma_{\rm register}}{\delta C_5 \delta C_5}\right|_{p\to 0} = \frac{1}{g_{6,\rm EFT}^2} \Delta_5,\qquad g_{6,\rm EFT}^2 = Z_{\rm match}\cdot g_{\rm reg}^2,$$

with the wave-function renormalization Z_{match} proved equal to 1. This is **not done here**.

> **F34.4 [DERIVED-CONDITIONAL on the register→EFT matching].** $g_{reg}^2 = 6$**A/Q** is the ZS-M6 register coupling; $g_{6,EFT}^2 = Z_{match} \cdot g_{reg}^2$. **Gate G-Charge(matching):** Z_{match} is OPEN; absent its proof, the continuum coupling is named g_{reg}, not g_6.

A separate object must not be conflated: **C_M^{sp} = 11 ln 2 + ln 3 = 8.7232** (ZS-S4) is the **Higgs-VEV** flat-direction determinant, *not* the gauge coupling [**AC17**].

5.2 Retraction of the even-dimensional torsion No-Go

v1.0 ruled out a "full-Y Ray–Singer torsion" source for the coupling/scale by asserting that the Y-sector is even-dimensional and even-dimensional analytic torsion vanishes. **This is retracted as a category error.**

1. The "6" of §2.1 is a **representation** dimension (dim $\Lambda^2(\mathbb{R}^{1,3})$), not the dimension of the manifold on which a torsion is defined. The torsion-relevant geometry in the corpus is the BCC T^3 quotient CW complex, not a 6-manifold.
2. Ray–Singer triviality is usually invoked for the full **unitary** de Rham complex. The corpus quantity is **not** that: F33's ln 4 is a **seam-parity-restricted determinant functional**, the determinant of the Laplacian restricted to the $\mathbb{Z}_2$-odd (seam-parity) sub-complex. The sign representation {±1} is itself orthogonal/unitary, so calling the bundle "non-unitary" is imprecise; the correct point is that the *restricted* complex is not the full de Rham complex, so the standard even-dimensional Hodge-duality cancellation does not apply to it. Concretely, F33.2B Computation ET finds this **2D T^2 seam-parity-restricted torsion to be ln 4 ≠ 0** [**AC14**] — even-dimensionality does *not* trivialize the restricted functional (and equivariant even-dimensional torsion has its own vanishing theorems, so "even ⇒ zero" was never the right lemma).
3. The spectral route to the *scale* is nonetheless closed-negative, but by the **F33.5/A28 involution-independent lattice no-go**: the achievable seam-parity-restricted log-determinants lie in $\{a \ln 2 + b \ln 3 : a, b \in \mathbb{Z}_{\geq 0}\}$, and the back-solved target 8.190 is not in that lattice [**AC15**].

> **F34.4-NG [RETRACTED → restated].** The even-dimensional-vanishing argument is withdrawn. The correct statements are: (i) the corpus's **seam-parity-restricted determinant torsion** (F33's terminology, retained here) is non-zero, ln 4 (AC14) — the issue is the restriction of the complex, not a failure of unitarity; (ii) the spectral route to the χ- *scale* is CLOSED-NEGATIVE by the lattice no-go (AC15), consistent with ZS-A28 keeping the absolute scale OPEN.

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§6. The Flux Multiplicity (F34.5)

This is the deepest of v1.0's overclaims, and v1.1 demotes it to an explicit open gate.

6.1 $b_2 = 1$ gives one flux; "83" is a trace, not a flux count

The geometric cycle has **$b_2(\Sigma_2) = 1$** [**AC18**], so §3 yields a **single** harmonic ω_2 and a single 4D four-form. The dark-energy "83" is *not* 83 two-cycles. In ZS-A24 the dark-energy sector is the projector P_Λ in the **type-II$_1$ corner** M_{obs}, carrying the **continuous trace** $\tau(P_\Lambda) = 83/121$; since $121 > Q = 11$, these are *not* integer projection ranks in M_{11} — they are continuous-trace weights [**AC19**, **AC20**]. So 83 is a **trace**, not a count of independent fluxes.

6.2 ZS-A29 is a kinematic precursor, not a proof

v1.0 wrote "ZS-A29 proves that this rank-83 flux vector is free." This **misstates A29**, which treats the rank-83 occupation only as a *Bousso–Polchinski-type kinematic precursor* and keeps **OPEN**: the G2 flux collectivization, whether the rank-83 occupation is a genuine independent flux vector, and the rank-to-energy identification. v1.1 carries these at A29's actual status.

6.3 The G-Flux decomposition: the theta-coupling quadratic form and its sub-gates (F34.M4)

v1.2 claimed to *settle* the realization of 83 as a flux coefficient; v1.3–v1.4 decomposed it but mislabelled the A_5 structure. v1.5 corrects the representation theory and states the robust object as the **theta-coupling quadratic form**, with the dimensionful charge factored out ($c_\theta = e_6\, \hat{c}_\theta$, $[\hat{c}_\theta] = 1$, $[e_6] = E^2$) and Z_{match} in its correct place:

$$\chi_- = \frac{1}{4\pi^2} \mathbf{c}_\theta^{\mathsf T} Z^{-1}\mathbf{c}_\theta = \frac{Z_{\rm match}\, g_{\rm reg}^2}{4\pi^2 V_\Sigma} e_6^2\, \hat{\mathbf c}_\theta^{\mathsf T} G^{-1}\hat{\mathbf c}_\theta,\qquad Z_{ab} = \frac{V_\Sigma}{Z_{\rm match}\, g_{\rm reg}^2} G_{ab},\quad G_{ab} = \tau(q_a K q_b).$$

(Defining $g^2_{6,EFT} = Z_{match}\, g^2_{reg}$ puts Z_{match} in the **denominator** of Z, so χ_- scales as $+Z_{match}$; v1.4 had this inverted [**GC12**].) Here **$\hat{\mathbf{c}}_\theta$** is the dimensionless **theta-coupling direction**, distinct from the membrane charge-lattice map Q_{source} (whose Smith-normal-form rank sets Path A/B); identifying them at rank 83 silently assumes equal θ-coupling ⇔ equal membrane charge.

(a) Coefficient-module monodromy. A flat rank-83 coefficient bundle carries commuting monodromies $U_\Lambda, V_\Lambda \in U(83)$; the massless count is $r_{flux} = \dim(\ker(U_\Lambda - I) \cap \ker(V_\Lambda - I))$, which for commuting unitaries equals the coinvariant dimension $\dim E/((U_\Lambda - I)E + (V_\Lambda - I)E)$ (lemma; verified both ways). The cases **r_{flux} = 83 (trivial), 82 (one phase), 0 (generic)** [**GC14**] make the *trivial* bundle a **load-bearing assumption**: motivated by the face algebra being a constant channel over Σ_2, but to be DERIVED one must compute $\rho_\Lambda(\gamma_1 = 2\pi i)$, $\rho_\Lambda(\gamma_2 = \log \lambda^*)$ on the actual face module.

(b) The 121 is an integral charge-lattice embedding (corrected in v1.4). v1.3 called the factor 121 a "basis ambiguity." That is wrong: a *genuine* basis change S sends $Z \to S^{-T} Z S^{-1}$ and $\mathbf{c}_\theta \to S^{-T}\mathbf{c}_\theta$, leaving **$\mathbf{c}_\theta^{T} Z^{-1}\mathbf{c}_\theta$ invariant** [**GC13**]. The factor 121 instead arises from re-declaring the **unit charge = 1** in two *different integral lattices* — the idempotent lattice ($\tau(q_a) = 1/121$) versus the orthonormal-τ lattice ($\hat{q}_a = \sqrt{121}\, q_a$) — i.e. it is an **integral charge-lattice embedding** ambiguity (two exhibited embeddings differ by 121; full index OPEN). The basis-independent quantity is $q_{min}^2 = \min_{v\in\Lambda} v^T G^{-1} v$. Fixing it is exactly the **finite-to-core lift F-A24.9**, now sharpened to a *trace-preserving integral charge-lattice intertwiner* $\iota_\Lambda : (\mathbb{Z}^{83}, Q_{fin}) \to (M_{obs}, \tau)$ with $\langle \iota e_a, \iota e_b\rangle_\tau = G_{ab}$ and $\iota(\Lambda_{fin}) = \Lambda_{core}$ — stronger than a Hilbert-space isometry, and **OPEN**.

(c) Kinetic centrality is OPEN, and A_5 alone does not close it (Schur corrected in v1.5). Centrality is the condition

$$P_\Lambda K P_\Lambda = k_\Lambda P_\Lambda .$$

Uniform trace does *not* give it: with $q_a = |a\rangle\langle a|$, $\tau(q_a K q_b) = (1/121) K_{aa} \delta_{ab}$, so orthogonality gives only **diagonality**; the counterexample $K = \mathrm{diag}(1, \ldots, 83)$ has equal face traces yet all-different Z_{aa} (a one-line consequence). v1.4 then claimed the corpus A_5 symmetry makes an invariant K "block-scalar with ≥4 distinct eigenvalues." **This misapplies Schur's lemma.** Because each irrep occurs with multiplicity, the dark-energy module is (for the illustrative baryon $6 = \mathbf{1} \oplus \mathbf{5}$)

$$H_\Lambda = (\mathbf{3}\otimes\mathbb{C}^4)\oplus(\mathbf{3'}\otimes\mathbb{C}^4)\oplus(\mathbf{4}\otimes\mathbb{C}^6)\oplus(\mathbf{5}\otimes\mathbb{C}^7),$$

and an A_5-invariant K is **not** four scalars but four **multiplicity matrices**,

$$K = \big(I_3\otimes K_3\big)\oplus\big(I_{3'}\otimes K_{3'}\big)\oplus\big(I_4\otimes K_4\big)\oplus\big(I_5\otimes K_5\big),\qquad K_3,K_{3'}\in M_4,\ K_4\in M_6,\ K_5\in M_7 .$$

The commutant has complex dimension $4^2 + 4^2 + 6^2 + 7^2 =$ **117** [**GC7**], so a Hermitian K carries up to 4 + 4 + 6 + 7 = **21** eigenvalues (and could, by coincidence, have as few as one — so "≥4 distinct" is also false). An explicit A_5-invariant but non-central example is $K = I_3 \otimes \mathrm{diag}(1, 2, 3, 4)$ on one sector [**GC8**]. The general A_5-invariant susceptibility is therefore

$$\chi_- = \frac{1}{4\pi^2}\sum_{\rho\in\{3,3',4,5\}} \mathbf{c}_\rho^{\dagger}\big(I_{d_\rho}\otimes K_\rho^{-1}\big)\mathbf{c}_\rho\qquad[\textbf{GC9}],$$

and the simple factor 83 requires *three* things at once — $K_\rho = k I$ for all ρ (scalarity), the same k across ρ (centrality), and the source alignment $\|c_\rho\|^2 \propto d_\rho m_\rho$. **Status: kinetic DIAGONALITY DERIVED-CONDITIONAL; kinetic CENTRALITY OPEN; A_5 alone is insufficient (commutant 117 > 1).**

(c′) The baryon embedding, the singlet, and the outer-automorphism selection (refined in v1.6). v1.5 observed that {baryon + dark energy} = $\mathrm{End}(V_{11}) - F_{TI}$ = $\mathbf{1{\cdot}1 \oplus 4{\cdot}3 \oplus 4{\cdot}3' \oplus 6{\cdot}4 \oplus 8{\cdot}5}$ [**GC3**] carries exactly one singlet, and that a nonzero A_5-invariant source needs dark energy to keep it. v1.6 makes this precise on two points a sixth review raised, then strengthens it.

First, the statement only makes representation-theoretic sense if **P_b and P_c are mutually orthogonal A_5-equivariant projectors** ([P_b, R(g)] = [P_c, R(g)] = 0); then P_Λ = I − P_b − P_c is A_5-invariant and the "location" of the singlet is well-defined. If P_b is not A_5-equivariant, the singlet has no definite location and one is already in the symmetry-broken branch (c).

Second, the selection does **not** uniquely pick 3 ⊕ 3′. Enumerating the rank-6 A_5-submodules of {baryon + DE} exhaustively [**GC17**, **GC18**]:

baryon 6	dark energy (83)	singlet in DE	invariant ĉ_θ
1 ⊕ 5	4·3 ⊕ 4·3′ ⊕ 6·4 ⊕ 7·5	0 [**GC4**]	none
2·3	1·1 ⊕ 2·3 ⊕ 4·3′ ⊕ 6·4 ⊕ 8·5	1	unique up to scale
3 ⊕ 3′	1·1 ⊕ 3·3 ⊕ 3·3′ ⊕ 6·4 ⊕ 8·5	1 [**GC5**]	unique up to scale
2·3′	1·1 ⊕ 4·3 ⊕ 2·3′ ⊕ 6·4 ⊕ 8·5	1	unique up to scale

> **Theorem F34.SEL (Singlet-Selection) [DERIVED-CONDITIONAL].** Conditional on P_b, P_c being mutually orthogonal A_5-equivariant projectors, a nonzero A_5-invariant linear theta-coupling exists **iff** the dark-energy module retains the single available singlet — i.e. iff baryon ≠ **1 ⊕ 5**. This *excludes* 1 ⊕ 5 but leaves three singlet-free candidates {2·3, 3 ⊕ 3′, 2·3′}; A_5 alone does **not** select 3 ⊕ 3′.

The extra structure that *does* select 3 ⊕ 3′ is the outer automorphism, and v1.7 **realizes it geometrically** rather than positing it. A_5 has $\mathrm{Out}(A_5) \cong \mathbb{Z}_2$, generated by σ (an odd permutation of the five points); σ exchanges the two 5-cycle classes, so on irreps **σ: 3 ↔ 3′, with 1, 4, 5 fixed** [**GC19a**], and among the singlet-free candidates 2·3 ↔ 2·3′ while **3 ⊕ 3′ is σ-stable** [**GC19b**]. The geometric content is the exterior square of the standard 4-dimensional module.

> **Theorem F34.BIV (Icosahedral Bivector Selection).** Let V_4 be the standard real four-dimensional A_5-module (the sum-zero subspace of the S_5 permutation representation on $\mathbb{R}^5$). Then $$\Lambda^2 V_4 \cong \mathbf 3 \oplus \mathbf 3',$$ a textbook character identity ($\chi_{\Lambda^2 4} = ½[\chi_4(g)^2 - \chi_4(g^2)] = (6, -2, 0, 1, 1) = \chi_3 + \chi_{3'}$, **GC27**). Over S_5 this is the six-dimensional *irreducible* (norm 1, **GC28**), so no S_5-equivariant splitting of the two triplets exists. In an oriented 4-dimensional **Euclidean** metric space (signature ++++, $\star^2 = +1$ on real 2-forms) $\Lambda^2 V_4 = \Lambda^2_+ \oplus \Lambda^2_-$ (self-dual ⊕ anti-self-dual, each 3-dimensional); $A_5 \subset SO(V_4)$ commutes with the Hodge star and **preserves each triplet**, while an orientation-reversing odd permutation (det = −1) anticommutes with the Hodge star and **swaps $\Lambda^2_+ \leftrightarrow \Lambda^2_-$**, i.e. **3 ↔ 3′** [**GC29**]. Thus 3 ↔ 3′ *is* self-dual ↔ anti-self-dual under

orientation reversal. The same $3 \oplus 3'$ is the standard decomposition of the six-dimensional crystallographic icosahedral representation $T_1 \oplus T_2$ (IMPORTED-PROVEN), an independent cross-check. **Status: PROVEN (representation theory).**

Signature caveat (v1.8). F34.BIV is formulated on the **Euclidean** standard A_5-module V_4, where $\star^2 = +1$ gives a *real* self-dual/anti-self-dual split. The corpus carrier Y = dim $\Lambda^2(\mathbb{R}^{1,3})$ is a **Lorentzian** bivector space, where on real 2-forms $\star^2 = -1$ and the self-dual/anti-self-dual decomposition appears only after complexification (as the $\pm i$ eigenspaces of $\star$). The identification of V_4's exterior square with the Lorentzian bivector carrier therefore requires complexification or an explicit signature-changing intertwiner, and remains part of the (DERIVED-CONDITIONAL) carrier bridge — not a real-orthogonal identity.

This converts v1.6's F34.OUT into two claims of *different* status, which v1.7 keeps separate:

F34.OUT-Math [DERIVED-CONDITIONAL]. Conditional on identifying the corpus rank-6 Y/baryon carrier with $\Lambda^2 V_4$, the module $\mathbf{3} \oplus \mathbf{3'}$ is selected with no additional representation assignment: it is the unique σ-stable singlet-free rank-6 A_5-module, and σ is realized as the orientation-reversing (Hodge ±-swapping) symmetry of F34.BIV. (The coincidence 6 = dim **X** × dim **Z** = 3 × 2 plays no role in the proof.) **Scope (v1.8):** F34.BIV selects the *representation type* $3 \oplus 3'$; it does **not** by itself fix the actual baryon projector P_b. Inside {baryon + dark energy} = $\mathbf{1} \oplus (3 \otimes \mathbb{C}^4) \oplus (3' \otimes \mathbb{C}^4) \oplus (4 \otimes \mathbb{C}^6) \oplus (5 \otimes \mathbb{C}^8)$, choosing a $3 \oplus 3'$ submodule means choosing one line in each multiplicity space $\mathbb{C}^4$, i.e. $P\_b = I_3 \otimes |u\rangle\langle u| \oplus I_3' \otimes |u'\rangle\langle u'|$ with $[u] \in \mathbb{C}P^3$; σ-stability forces $u' = S\_\sigma\, u$ but a residual $\mathbb{C}P^3$ multiplicity-line freedom remains. Fixing P_b is a *dynamical* question (a parent-Hessian spectral selection), deferred to ZS-F35 (§10).

G-Outer-Physical [OPEN]. Whether the orientation-reversing σ is a *physical* Z-Spin symmetry — seam-orientation reversal, J_Z parity, a register automorphism, or a CPT-related exchange — is open. F34.BIV makes σ geometric (a Hodge involution), which sharpens but does not close this identification.

The carrier bridge is the more honest framing of the earlier "Y = bivector module": the corpus already uses Y = 6 = dim $\Lambda^2(\mathbb{R}^{1,3})$, and F34.BIV says the *internal* rank-6 module is likewise an exterior square — of the icosahedral V_4.

Three source-symmetry paths then follow [**GC10**, **ID1**]:

- **(a) dark energy keeps the singlet, K is A_5-invariant.** Then $[K, R(g)] = 0$ forces the singlet block to decouple, $P\_s\, K\, (I - P\_s) = 0$ [**GC21**], and an A_5-invariant source couples only to the **1-dimensional** invariant subspace. The susceptibility is a **single mode** (see §7), and the factor 83 is *replaced*. (If K breaks A_5, even a singlet source mixes via the Schur complement [**GC22**],

$(Z^{-1})\_ss = 1/(Z\_ss - B D^{-1} B\dagger) \neq 1/Z\_ss$ — so the kinetic A_5-invariance is a genuine extra hypothesis.)

- **(b) dark energy has no singlet** (baryon = 1 ⊕ 5). An exact-A_5 linear source must vanish; a nonzero source needs explicit/spontaneous A_5 breaking or a spurion ĉ_θ = I(s).
- **(c) A_5 is broken in the source sector.** Then ĉ_θ is a spurion and the factor 83 returns *only* with the alignment ‖c_ρ‖² ∝ d_ρ m_ρ together with full centrality. **The central number 0.091847 lives entirely in path (c).**

So the honest content is a *trichotomy*; single-mode (a) is **one branch, not generic**, since the singlet's location is open. Determining the branch requires constructing the actual baryon/CDM projectors and the vacuum character (§10).

(d) The theta-coupling direction is resolved by (c′): in path (a) ĉ_θ is the singlet (fixed up to scale, *not* uniform in the face basis); the uniform ĉ_θ = (1, …, 1) belongs to path (c). A single θ-mode does **not** imply rank Q_source = 1 — the susceptibility direction ĉ_θ and the membrane charge-lattice map Q_source stay distinct, so Path A/B is still set by rank Q_source.

(e) Path A vs Path B. Set by the Smith normal form of Q_source ∈ M_{N×r_flux}(ℤ): rank r_flux → Path A, rank 1 → Path B, intermediate → partial; elementary divisors d_i = 1 → primitive. The v1.2 claim that orthogonality excludes Path B is **withdrawn**. Status: **Path selection OPEN.**

6.4 Honest status: decomposed, not closed

> **F34.M4 [DECOMPOSITION; centrality OPEN; A_5 paths computed].** The flux coefficient decomposes into {monodromy (a), charge-lattice embedding (b), kinetic centrality (c), A_5 source paths (c′), Path selection (e)}. Only kinetic *diagonality* is derived; **centrality is OPEN and A_5 alone cannot close it** — the commutant is $M_4 \oplus M_4 \oplus M_6 \oplus M_7$ (dim 117), permitting up to 21 eigenvalues, not four scalars. The source structure resolves into a **trichotomy** (F34.SEL): (a) single-mode if dark energy keeps the lone singlet *and* K is A_5-invariant; (b) zero unless A_5 is broken, if it does not; (c) the factor-83 form only under A_5-breaking with source alignment. The baryon module is selected uniquely to 3 ⊕ 3′ only with the outer automorphism (F34.OUT). The 121 is an integral-lattice embedding (two embeddings differ by 121; full index OPEN), and Path A/B awaits the Smith normal form of Q_source. **Gate G-Flux: DECOMPOSED; the central factor-83 candidate lives only in path (c); path (a) admits the single-mode closure of §10.**

The Bousso–Polchinski *gap* (dense discretuum) remains a separate, still-open question (the shell density of Λ(**n**) = Λ_bare + ½ **n**^T Q **n** needs Q fixed and counted; §10); the susceptibility does not require it.

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§7. The Susceptibility: the Finite-Symmetry Rank Theorem (F34.SR, F34.6)

The single most transferable statement in this paper is that the source-coupled susceptibility of a multi-four-form theory is controlled not by the number of fluxes but by the dimension of the invariant subspace the source lives in.

> **Theorem F34.SR (Finite-Symmetry Susceptibility-Rank) [PROVEN].** Let a finite group G act unitarily on a finite-dimensional flux space V, let $Z > 0$ be G-equivariant ($R(g)^\dagger Z R(g) = Z$ for all g), and let the source satisfy $c \in V^G$ ($R(g)c = c$). Then Z^{-1} is G-equivariant and preserves the orthogonal decomposition $V = V^G \oplus (V^G)^\perp$, so the off-block part of Z^{-1} annihilates c and $$\chi = \frac{1}{4\pi^2}\mathbf c^{\dagger} Z^{-1}\mathbf c = \frac{1}{4\pi^2}\mathbf c^{\dagger}\big(Z|_{V^G}\big)^{-1}\mathbf c,\qquad N_{\rm eff} = \dim V^G.$$ The effective mode count is the **invariant-source rank $\dim V^G$, not the full flux rank $\dim V$** [**GC30**]. More precisely, if the admissible sources occupy a subspace $S_{src} \subseteq V^G$, the response is confined to V^G and the maximal invariant-source rank is $N_{eff} = \dim S_{src} \leq \dim V^G$, with equality when the admissible sources span V^G. Corollaries: **(i)** $\dim V^G = 0 \Rightarrow c = 0$ (no invariant source); **(ii)** $\dim V^G = 1 \Rightarrow$ an exact **single-mode** susceptibility; **(iii)** $\dim V^G = m > 1 \Rightarrow$ an $m\times m$ invariant-block susceptibility $c^\dagger(Z|_{V^G})^{-1}c$. (In F34's exact-$A_5$ branch $\dim V^{A_5} = 1$, so $S_{src} = V^{A_5}$ and $N_{eff} = 1$ regardless.)

The proof is one line of representation theory (a G-equivariant operator preserves isotypic components, and V^G is the trivial-isotypic one), but the consequence is not folklore in the flux-landscape setting: the common heuristic "N fluxes contribute N modes to the vacuum susceptibility" is **false** whenever the source respects a symmetry — only the invariant directions respond. Changing the kinetic eigenvalues on $(V^G)^\perp$ leaves χ untouched [**GC31**]. In F34 this is exactly why the "factor 83" was never the right multiplicity for an A_5-invariant source.

Application to F34. With the master normalization ($c_\theta = e_6 \hat{c}_\theta$, $[\hat{c}_\theta] = 1$, $[e_6] = E^2$),

$$\chi_- = \frac{1}{4\pi^2}\mathbf{c}_\theta^{\mathsf T} Z^{-1}\mathbf{c}_\theta = \frac{Z_{\rm match}g_{\rm reg}^2}{4\pi^2V_\Sigma}e_6^2\hat{\mathbf c}_\theta^{\mathsf T} G^{-1}\hat{\mathbf c}_\theta,\qquad Z = \frac{V_\Sigma}{Z_{\rm match}g_{\rm reg}^2}G,\quad G_{ab} = \tau(q_a K q_b),$$

with Z_{match} in the **denominator** of Z, hence the **numerator** of χ_- [**GC12**]. In the baryon/CDM-equivariant single-singlet branch, $\dim H_\Lambda^{A_5} = 1$, so by F34.SR $N_{eff} = 1$ and the susceptibility is single-mode — *without* proving full 83-dimensional centrality. Which number χ_- yields depends on the A_5 source path of §6.3(c′):

- **Path (a) — dark energy keeps the singlet, K is A_5-invariant ($\dim V^G = 1$).** A_5 fixes the singlet *direction* s but not its *magnitude*. Writing the primitive integral source as $v_s = \nu_s s$, with the **lattice norm** $\nu_s^2 := v_s^{\mathsf T} v_s = \langle v_s, v_s\rangle_0$ and the **Hessian norm** $G_s := s^{\mathsf T} G s$, the full quadratic form is $q_s := v_s^{\mathsf T} G^{-1} v_s = \nu_s^2/G_s$ (since for A_5-invariant G the singlet is an eigenvector, $G^{-1}s = G_s^{-1}s$), so

$$\chi_-^{(s)} = \frac{Z_{\rm match}g_{\rm reg}^2}{4\pi^2V_\Sigma}e_6^2\frac{\nu_s^2}{G_s} = \frac{Z_{\rm match}g_{\rm reg}^2}{4\pi^2V_\Sigma}e_6^2q_s\qquad[\textbf{GC23}, \textbf{GC32}].$$

(**v1.7 fix:** v1.6 wrote $\nu_s^2 = v_s^{\mathsf T} G^{-1} v_s$ *and* divided by G_s, double-counting the kinetic norm; the corrected reading separates the lattice norm $\nu_s^2 = v_s^{\mathsf T} v_s$ from $G_s = s^{\mathsf T} G s$.) There is no factor 83 — but the integral-lattice (121 / F-A24.9) problem **persists as the single number ν_s**. It is convenient to bundle the three remaining single-mode unknowns into one basis-independent **Singlet Response Invariant**

$$\boxed{I_s := Z_{\rm match}v_s^{\mathsf T} G^{-1} v_s}\qquad\Longrightarrow\qquad \chi_-^{(s)} = \frac{g_{\rm reg}^2e_6^2}{4\pi^2V_\Sigma}I_s,$$

which folds together register-to-EFT matching ($Z_{\rm match}$), the singlet Hessian norm (G_s), and the primitive-source lattice normalization (ν_s). **What F34 asks of an external calculator is therefore not the full 83×83 matrix but a single zero-momentum singlet-response residue I_s** (definition basis-independent: DERIVED; value: OPEN, → ZS-F35). (If K breaks A_5, even this is corrected by the Schur complement $(Z^{-1})_{ss} = 1/(Z_{ss} - B D^{-1} B^\dagger)$ [**GC22**].)

- **Path (b) — no singlet ($\dim V^G = 0$).** An exact-A_5 source vanishes; a nonzero source needs breaking or a spurion.
- **Path (c) — A_5 broken, fully aligned and central.** Only here does the factor 83 appear. Setting $V_\Sigma = 1$, $G = I$ (central), $\hat c_\theta = (1, \dots, 1)$, $Z_{\rm match} = 1$,

$$\chi_- = \frac{83 g_{\rm reg}^2}{4\pi^2} e_6^2 = 0.091847\cdot e_6^2\qquad[\textbf{GC11}],\qquad \rho_{\Lambda,Z} = \tfrac12\chi_-\omega^2 = 0.234402\cdot e_6^2\qquad[\textbf{GC16}].$$

(The v1.3 value $\rho = 0.234426$ was an arithmetic slip; $0.09184669 \times \omega^2/2 = 0.09184669 \times 2.5521043 =$ **0.23440234**.) The shift from the v1.2 number 0.066270 to 0.091847 is the metric correction V_Σ : 1.385949 → 1 (a factor $2\pi\,\kappa_\lambda = 0.7215$).

> **F34.6 [CANDIDATE — path (c), aligned].** $\chi_- = 0.091847 \cdot e_6^2$ (with $Z_{\rm match} = 1$) is the value of the master form **only** in path (c) at the fully aligned central point {$V_\Sigma = 1$; $G = I$ (central — A_5 alone does not give this, commutant 117); $\hat c_\theta = (1, \dots, 1)$ with the alignment $\|c_\rho\|^2 \propto d_\rho m_\rho$; orthonormal-τ lattice (F-A24.9); $Z_{\rm match} = 1$; anomaly-free}. In the exact-A_5 single-mode branch (a) the susceptibility is instead $(Z_{\rm match}\, g_{\rm reg}^2/4\pi^2\, V_\Sigma)\, e_6^2$

$q_s = (g_{reg}^2 e_6^2/4\pi^2 V_\Sigma) I_s$, controlled by the single invariant I_s, with **no** factor 83. The only basis-independent statement is the master form itself.

Corollary F34.SR-Branch (Symmetry-Branch Ratio). At fixed $\{V_\Sigma, Z_{match}, e_6\}$, the two branches differ by a *structural* factor, not a normalization: $$\frac{\chi_{83}^{(c,\rm aligned)}}{\chi_{\rm singlet}^{(a)}} = \frac{83}{q_s}.$$ At the benchmark $q_s = 1$ the ratio is exactly **83**. Matching the same observed vacuum density in the single-mode branch instead of the aligned-83 branch raises the characteristic charge scale by $83^{1/4}$: where path (c) reports $e_6 \approx (3.219\ \text{meV})^2$, path (a) at $q_s = 1$ reports $\sqrt{e_6} \approx 9.72$ meV. This is **not** a prediction; it is a conditional discriminator that an independent determination of e_6 (ZS-F35) would resolve. **Status: DERIVED-CONDITIONAL / REPORTED.**

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§8. The Residual: One Dimensionful Charge Behind a Conditional List (F34.8)

The cleanest way to record the residual is to write the master form and label which gate sits in each factor (the under-labels are given in prose to avoid fragile markup):

$$\chi_- = \frac{1}{4\pi^2} \mathbf{c}_\theta^{\mathsf T} Z^{-1}\mathbf{c}_\theta,\qquad Z = \frac{V_\Sigma}{Z_{\rm match}\, g_{\rm reg}^2} G,\qquad G_{ab} = \tau(q_a K q_b),\qquad \mathbf{c}_\theta = e_6\, \hat{\mathbf c}_\theta .$$

The factor V_Σ carries G-Metric; the prefactor Z_{match}/g_{reg}^2 carries G-Charge; the matrix $G = \tau(q_a K q_b)$ carries the G-Flux sub-gates (a) monodromy, (b) charge-lattice embedding, and (c) centrality; the direction $\hat{c}_\theta$ carries the G-Flux source paths (c′); and rank Q_{source} carries the Path-A/B sub-gate (e).

Gate	Factor it controls	v1.7 status
G-Metric	V_Σ	**Candidate normalization** $V_\Sigma = 1$ (principal-polarization representative; F34.M3). Koenigs RETRACTED. METRIC-IDENTIFICATION-CONDITION.
G-Flux (a)	monodromy r_{flux}	Trivial bundle ($r = 83$) a load-bearing assumption (toy holonomy only).
G-Flux (b)	charge-lattice embedding	Two exhibited embeddings differ by 121; full index OPEN → F-A24.9.
G-Flux (c)	$G = \tau(q_a K q_b)$	Diagonality DERIVED; **centrality OPEN**; A_5 commutant **117** ($M_4 \oplus M_4 \oplus M_6 \oplus M_7$), not four scalars.

Gate	Factor it controls	v1.7 status
G-Flux (c′)	$\hat{c}\_\theta$ (source)	**Trichotomy** (F34.SEL): (a) single-mode singlet, (b) zero unless A_5 broken, (c) factor 83 under alignment.
G-Flux (e)	rank Q_source	Path A/B OPEN (Smith normal form).
G-Charge	Z_match, e_6	Z_match (denominator of Z) OPEN; $\mathbf{e_6}$ the sole *dimensionful* OPEN (F33-G12).

The dimensionful content does reduce to the **single charge $\mathbf{e_6}$**, but only *behind* this list. Were the list discharged in path (c) at the aligned central point and e_6 fixed, reproducing ρ_Λ,obs ≈ $(2.24\ \text{meV})^4$ would require $\mathbf{e_6 \approx (3.219\ meV)^2}$ [REPORTED, not a claim; firewall F34-G5].

> **F34.8 [OPEN — the honest terminus].** The robust result is the master form $\chi_- = (Z\_match\ g\_reg^2/4\pi^2\ V\_\Sigma)\ e_6^2\ \hat{c}\_\theta^T\ G^{-1}\ \hat{c}\_\theta$. The dimensionful residual is e_6, but the dimensionless coefficient is conditional on the factor map above, of which only the metric *class* ($c_1 = 1$), kinetic *diagonality*, and the *representation-theoretic structure* (the singlet count, the commutant 117, the outer-automorphism σ-stability of $3 \oplus 3'$) are settled — kinetic *centrality*, the singlet's location, the charge-lattice embedding (now ν_s in the single-mode branch), Path selection, anomaly cancellation, and Z_match are not. "Sole remaining dimensionful OPEN = e_6" may be written **only after** the list is discharged; v1.7 does not claim it. The contribution is the **bivector selection** of the baryon module (F34.BIV), the **finite-symmetry susceptibility-rank theorem** (F34.SR, N_eff = dim V^G), and the **single-mode reduction** — by F34.SR the 83-dimensional flux/source realization reduces to the five one-dimensional objects {P_s, G_s, ν_s, (U_s, V_s) = 1, Q_s}, while the **independent G-Metric (V_Σ) and G-Charge (Z_match, $\mathbf{e_6}$) gates remain**. This is a *reduction*, not a closure; the actual P_b, I_s, e_6 are deferred to ZS-F35.

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§9. Falsification Gates

Gate	Layer	Condition that voids the result
G-A1	Mathematical	$\Sigma_2 \neq E\_\lambda^*$, or a 6D spacetime metric is asserted that the corpus does not supply.
G-A2	Theoretical	Any anomaly gate fails (lattice non-unimodular, tadpole/global-anomaly non-cancelling, nucleation ≠ one unit).
G-Metric	Mathematical	V_Σ claimed *derived* (it is a metric-identification candidate); or the Koenigs argument re-used (RETRACTED); or V_Σ = 2 ln 2 claimed.

Gate	Layer	Condition that voids the result
G-Flux	Mathematical	Kinetic *centrality* claimed from uniform trace or from A_5 alone (commutant is 117, not $\mathbb{C}I$); an A_5-invariant K called "block-scalar" or "≥4 distinct eigenvalues" (it is four multiplicity matrices, ≤ 21 eigenvalues); the factor 83 used without specifying path (c) + alignment; or 0.091847 called derived.
G-Source	Mathematical	A nonzero A_5-invariant linear ĉ_θ asserted when dark energy has no singlet; the single-mode result (path a) stated without the kinetic A_5-invariance hypothesis ($[K, R(g)] = 0$) or with 1/Z_ss in place of the Schur complement when A_5 is broken; or the single-mode ν_s normalization dropped (with $\nu_s^2 = v_s^T v_s$ and $G_s = s^T G\, s$, **not** $\nu_s^2 = v_s^T G^{-1} v_s$, which double-divides — the v1.6 error); or the singlet holonomy written U_s = V_s rather than the correct gate U_s = V_s = **1** (U_s = V_s = −1 is also equal yet kills the invariant singlet).
G-Outer-Math	Mathematical	$\Lambda^2 V_4 \neq 3 \oplus 3'$ asserted (it is a character identity, GC27); the Hodge ±-swap under odd permutations denied (GC29); or baryon = $3 \oplus 3'$ asserted as *uniquely selected* without the σ-stability hypothesis (A_5 alone leaves $\{2{\cdot}3, 3 \oplus 3', 2{\cdot}3'\}$).
G-Outer-Physical	Theoretical	F34.OUT-Math (a DERIVED-CONDITIONAL representation statement) presented as a physical derivation — the identification of the orientation-reversing σ with a Z-Spin symmetry is OPEN; or "favours $3 \oplus 3'$" justified by $6 = 3 \times 2$ alone.
G-SuscRank	Mathematical	the susceptibility's effective mode count equated with the full flux rank dim V rather than dim V^G (F34.SR, GC30); or the m > 1 invariant block reduced to a scalar without computing Z|_{V^G}.
G-Generic	Consistency	The single-mode branch (a) called "generic" — the singlet's location (baryon vs DE) is open, so no branch is generic.
G-Carrier	Mathematical	$M_4 \times \Sigma_2$ claimed the *unique* carrier without a minimality/completeness principle (degree counting gives dim ≥ 6 only); or six observable spacetime dimensions asserted.
G-Charge	Meta / matching	g_6 used as the continuum coupling without Z_match; Z_match placed in the numerator of Z; e_6 back-solved from observed ρ_Λ.
G-Torsion	Mathematical	An even-dimensional-vanishing argument is re-used (category error); or the scale claimed from a determinant lying outside {a ln2 + b ln3}.

Gate	Layer	Condition that voids the result
G-Sym	Consistency	ω_KV (= √2) conflated with arg λ^* (= 2.2592).

Layering: **G-A1, G-Flux, G-Torsion** are theoretical-collapse gates; **G-A2, G-Metric, G-Sym** are consistency/computation gates; **G-Charge** is the meta / anti-numerology gate.

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§10. Outlook

Thermal history (moved from v1.0 §F34.7). The dynamical evolution of ρ_Λ is the Klinkhamer–Volovik q-theory with $c_{KV} = 1/8$ and the freeze-out kernel $\dot\rho_V = -\Gamma_Y(\rho_V - \rho_{V,0})$, where Γ_Y is the **Y-sector thermal** correlator (the SM plasma thermalizes), *not* the Z-Goldstone — so the apparent ZS-M43 No-Thermal-Transport conflict does not arise (M43 retires only the unitary Z-Goldstone η/s). This is a *consequence* of the framework, not an input to the charge-unit reduction, and is recorded here as context.

A pre-registered programme to close the gates.

1. **G-Metric (metric identification):** derive the internal Hodge metric as the parent-action kinetic Hessian $\delta^2 S_{parent} = \int_{\Sigma_2} \delta\omega_2 \wedge \star_{g_{Hess}} \delta\omega_2$ and check $g_{Hess} = g_\Theta = |dz|^2/\mathrm{Im}\,\tau$ (Area = 1). Equality lifts F34.M3 to DERIVED; proportionality $g_{Hess} = \alpha\, g_\Theta$ exposes α as a new normalization gate.
2. **Baryon embedding + outer automorphism (decides the branch).** Construct the actual projectors, not the decomposition: build P_c (CDM), P_b (baryon), $P_\Lambda = I - P_b - P_c$ on $V_{11} \otimes V_{11}^*$ (e.g. via the central projectors $P_\rho = (d_\rho/60)\, \Sigma_g\, \overline{\chi_\rho(g)}\, R(g)$ plus matrix-unit projectors within each isotypic block), verify $P_i^2 = P_i = P_i^\dagger$, $P_b P_c = 0$, $[P_b, R(g)] = [P_c, R(g)] = 0$, and read the multiplicities from $\chi_\Lambda(g) = \mathrm{Tr}(P_\Lambda R(g))$. This decides the singlet's location (F34.SEL). F34.BIV already realizes the selecting symmetry geometrically (the Hodge ±-swap of $\Lambda^2 V_4$, GC29); what remains (G-Outer-Physical) is to identify that orientation-reversing involution with a *physical* Z-Spin symmetry and prove $\sigma P_b \sigma^{-1} = P_b$ — the candidates to test are seam-orientation reversal, J_Z parity, a register automorphism, and a CPT-related exchange; success upgrades G-Outer-Physical from OPEN to DERIVED, completing the bridge to F34.BIV.
3. **Single-mode reduction of path (a) — the efficient route.** If step 2 leaves the singlet in dark energy and K is A_5-invariant, the susceptibility is the single mode of §7 and, by F34.SR, the **83-dimensional flux/source realization** reduces to **five one-dimensional objects** — no 83-dimensional centrality required (the **independent G-Metric (V_Σ) and G-Charge (Z_{match}, e_6) gates still remain**): (i) the rank-1 singlet projector $P_s = (1/60)\, \Sigma_g\, R_\Lambda(g)$; (ii) the singlet Hessian norm $G_s = \tau(P_s K_{Hess} P_s)/\tau(P_s)$ from the parent Hessian projected to the singlet (needing only $[P_s, K_{Hess}] = 0$ and $G_s > 0$, not full centrality); (iii) the primitive integral singlet $v_s = (\Sigma_g R(g) v)/\gcd$, with lattice norm $\nu_s^2 = v_s^T v_s$ and quadratic form q_s =

$v_s^{T} G^{-1} v_s = \nu_s^2/G_s$ (the single-mode version of the 121 / F-A24.9 problem) — bundled with Z_{match} into I_s; (iv) the singlet holonomy $(U_s, V_s) = (P_s U_\Lambda P_s, P_s V_\Lambda P_s)$, requiring the gate **$U_s = V_s = 1$** ($r_{singlet} = 1$, not $r_{flux} = 83$; $U_s = V_s = -1$ is equal yet kills the singlet); (v) the singlet nucleation $Q_s = Q_{source} P_s$, whose Smith normal form d_s decides primitive one-unit nucleation. Full 83-dimensional centrality (the double-commutant test below) and the factor 83 are needed **only in path (c)**. A key simplification, to be proved in ZS-F35, decouples this route from step 2: since {baryon + dark energy} carries **exactly one** singlet and any singlet-free A_5-equivariant baryon projector satisfies $P_s P_b = 0$ (hence $P_s P_\Lambda = P_s$), the unique dark-energy singlet projector P_s — and therefore I_s — is **independent of the residual $\mathbb{C}P^3$ multiplicity-line choice** in P_b (a *Baryon-Embedding Independence* statement). I_s can thus be computed without first closing P_b. **These computations are deferred to ZS-F35** (see below); F34 terminates with the reduction in place.

4. **Path (c) only — full centrality.** Since the A_5 commutant is 117-dimensional, A_5 alone cannot centralize K; either enlarge the symmetry to $G_\Lambda = \mathrm{Aut}(A_{face}, P_b, P_c, \tau, Q_{source})$ and test $\mathrm{End}_{G_\Lambda}(H_\Lambda) = \mathbb{C}I$ (a double-commutant test, strictly stronger than A_5-invariance), or compute the full parent-Hessian K_{ab}, restrict to P_Λ, extract the multiplicity matrices K_ρ, and measure $\Delta_{cent} = \|P_\Lambda K P_\Lambda - \bar{k} P_\Lambda\|_F / \|P_\Lambda K P_\Lambda\|_F$ (centrality iff $\Delta_{cent} = 0$; isotypic-scalar gives the four-block $\chi_- = (1/4\pi^2) \Sigma_\rho k_\rho^{-1}\|c_\rho\|^2$; generic keeps the full matrix form).
5. **G-Carrier:** state and justify a Minimal-Carrier / Corpus-Completeness principle to upgrade $M_4 \times \Sigma_2$ from minimal to unique.
6. **The two follow-on papers (pre-registered split).** The residual quantities have *different logical characters* — P_b is a representation/projector-selection problem, I_s is a dimensionless zero-momentum matching problem, and e_6 is a dimensionful UV-charge problem that (per F33's Charge-Unit Obstruction) cannot be derived without a new dimensionful input — so they are split across two papers rather than forced into one. **ZS-F35** (*The Universal Exact-A_5 Singlet Lift*) closes the dimensionless and lattice content: it proves the **Multiplicity-Line No-Go** (A_5 + σ fix the type $3 \oplus 3'$ but leave a $\mathbb{C}P^3$ projector family), the **Baryon-Embedding Independence Theorem** ($P_s P_b = 0$, $P_s P_\Lambda = P_s$), a direct **physical singlet stiffness** $Z_s^{phys} := \delta^2\Gamma_{EFT}/\delta F_s \delta F_s|_{p=0} = V_\Sigma G_s/(Z_{match} g_{reg}^2)$ bundling $\{V_\Sigma, Z_{match}, G_s\}$ into one EFT coefficient so that $\chi_-^{(s)} = e_s^2/(4\pi^2 Z_s^{phys})$ with $e_s = e_6 \nu_s$, the **primitive singlet lift** (ν_s, $U_s = V_s = 1$, $d_s = \gcd(Q_{source} v_s)$), and an optional parent-Hessian spectral selection of the actual P_b (with the action-level justification that the *minimal* Hessian mode is the baryon, not a value-fitted eigenvector); it may also include an **Approximate-Symmetry Stability Theorem** showing the single mode is *second*-order stable under weak kinetic A_5-breaking, $(Z^{-1})_{ss} = 1/(Z_s - \varepsilon^2 B D^{-1} B^\dagger) \Longrightarrow \delta\chi_s/\chi_s = O(\varepsilon^2)$. **ZS-F36** (*The Primitive Membrane Charge and the Absolute Z-Spin Vacuum Scale*) then closes the sole dimensionful unknown e_6 from a *pre-registered* UV completion — either an explicit wrapped-brane charge $e_6 = \mu_4$ (with flux quantization, tadpole/global-anomaly cancellation) or dimensional transmutation $e_6 = c_e \Lambda_-^{2}$ from a specified odd gauge group G_- (matter content, b_-, M_{UV}, and c_e all fixed *before* computing, to avoid numerology). Only after F36 may the absolute scale be claimed.

§11. Conclusion

ZS-F34 has been through seven external reviews, each catching the previous version's over-claim. v1.1 named three gates; v1.2 claimed to close two; v1.3 retracted those; v1.4 corrected the centrality and carrier errors; v1.5 corrected the A_5 representation theory and computed a source trichotomy; v1.6 split off the outer-automorphism selection and the single-mode reduction; **v1.7 (this terminal version) fixes one real formula error and places two general theorems at the front.** The version-by-version growth was not endless polishing but the steady addition of load-bearing claims, each bringing a new verification target; v1.7 adds the last two and stops.

Two general theorems. (i) **F34.BIV (Icosahedral Bivector Selection).** The exterior square of the standard 4-dimensional A_5-module is $\Lambda^2 V_4 = \mathbf{3} \oplus \mathbf{3'}$ (character, PROVEN); it is the 6-dimensional S_5-irreducible, and the two triplets are the Hodge self-dual / anti-self-dual sectors, *swapped* by an orientation-reversing odd permutation. This **realizes** v1.6's outer automorphism geometrically — $3 \leftrightarrow 3'$ is self-dual ↔ anti-self-dual under orientation reversal — and is cross-checked by the known $T_1 \oplus T_2$ decomposition of the 6-dimensional crystallographic icosahedral representation. Conditional on identifying the corpus rank-6 Y/baryon carrier with $\Lambda^2 V_4$, the module $3 \oplus 3'$ is selected (F34.OUT-Math, DERIVED-CONDITIONAL); the physical identity of the orientation-reversing symmetry stays OPEN (G-Outer-Physical). (ii) **F34.SR (Finite-Symmetry Susceptibility-Rank).** For a G-equivariant kinetic operator and a source in V^G, $\chi = (1/4\pi^2)\, c^\dagger (Z|_{V^G})^{-1} c$, so the effective mode count is $\dim V^G$, not the full flux rank. The single-mode dark-energy result and the m×m block case are corollaries; the "factor 83" was never the right multiplicity for an invariant source. This is a statement about any finite-symmetry multi-four-form theory, valid outside Z-Spin.

One error fixed. The single-mode susceptibility is $\chi_- = (Z_{match}\, g_{reg}^2/4\pi^2 V_\Sigma)\, e_6^2\, (\nu_s^2/G_s)$ with $\nu_s^2 = v_s^T v_s$ (lattice norm) and $G_s = s^T G s$ (Hessian norm), since $q_s = v_s^T G^{-1} v_s = \nu_s^2/G_s$ already; v1.6 divided by G_s twice. The three single-mode unknowns bundle into one basis-independent **Singlet Response Invariant** $I_s = Z_{match}\, v_s^T G^{-1} v_s$.

The terminus. The robust deliverable is the master form $\chi_- = (Z_{match}\, g_{reg}^2/4\pi^2 V_\Sigma)\, e_6^2\, \hat{c}_\theta^T G^{-1} \hat{c}_\theta$; F34.SR fixes its effective rank at $\dim V^G$, F34.BIV selects the rank-6 carrier, and the exact-A_5 branch reduces to I_s plus the **independent** G-Metric (V_Σ) and G-Charge (Z_{match}, e_6) gates. Single-mode is **one of three branches**, not generic — the singlet's location is open. The actual P_b, I_s, and e_6 are deferred to **ZS-F35** (*The Exact-A_5 Singlet Lift and the Primitive Dark-Energy Charge*). The honest contribution of v1.7 is the reframing of F34 from a conditional internal coefficient into a *symmetry-reduction theory of multi-top-form vacuum susceptibility* — neither a forced closure nor a No-Go, consistent with the corpus's standing caution that internal iteration converges to honesty, not to closure. **F34 terminates here.** (**A**, **Q**, dim **Z**) = (35/437, 11, 2) **LOCKED**.

Acknowledgements & Code Availability

Seven external reviews have shaped this paper. v1.1 reframed v1.0's over-claims as three gates; v1.2 claimed to close two; v1.3 retracted that (Koenigs non sequitur; hidden 121); v1.4 corrected the centrality and carrier errors and introduced an A_5 analysis; v1.5 corrected that analysis (Schur is four *multiplicity matrices*, commutant 117, not four scalars; the 83-decomposition is baryon-conditional; Z_match was inverted) and computed a source trichotomy instead of a No-Go; v1.6 corrected three v1.5 over-reaches (singlet-selection only *excludes* $1 \oplus 5$ and needs A_5-equivariant projectors; the single-mode reduction needs the kinetic operator A_5-invariant; the singlet normalization ν_s is not fixed by A_5) and proposed an outer-automorphism selection plus a single-mode reduction. A **seventh** review found one real formula error — the ν_s^2/G_s double-counting ($v_s^T G^{-1} v_s$ already equals ν_s^2/G_s) — fixed here, and recommended terminating F34; a parallel deep exploration showed the result's value is raised most by stating two general theorems rather than closing another gate. v1.7 does both. The two theorems are: the **Icosahedral Bivector Selection Theorem** (F34.BIV — $\Lambda^2 V_4 = 3 \oplus 3'$, the two triplets being the Hodge self-dual/anti-self-dual sectors swapped by an orientation-reversing odd permutation, which *geometrizes* the outer automorphism; the carrier identification is DERIVED-CONDITIONAL and the physical σ stays OPEN), and the **Finite-Symmetry Susceptibility-Rank Theorem** (F34.SR — for G-equivariant $Z > 0$ and source $c \in V^G$, $\chi = (1/4\pi^2)\, c^\dagger (Z|_{V^G})^{-1} c$, so $N_{eff} = \dim V^G$, not the full flux rank; valid for any finite-symmetry multi-four-form theory). The single-mode unknowns are bundled into the basis-independent Singlet Response Invariant $I_s = Z_{match}\, v_s^T G^{-1} v_s$. Two suites accompany the paper: zs_f34_verify_v1_1.py (23/23, arithmetic and the v1.1 regressions) and zs_f34_verify_v1_8.py (32/32, GATE-DECOMPOSITION-v5 — arithmetic, counterexample, and model-instantiation checks, **not** a closure certificate), which keeps the v5 checks (rank-6 enumeration and singlet counts, the character-level outer automorphism, the explicit A_5 5-point representation, the Schur-complement mixing, and — now corrected — the single-mode normalization with $\nu_s^2 = v_s^T v_s$ and $G_s = s^T G\, s$) and adds GC27–GC32: the $\Lambda^2(4) = 3 \oplus 3'$ exterior-square character (GC27), its irreducibility as the 6-dim S_5 rep (GC28), the even-commutes/odd-anticommutes Hodge-star check that realizes σ (GC29), the F34.SR invariant-block reduction at $\dim V^G = 2$ (GC30), the invariance of the single mode under non-singlet kinetic changes (GC31), and the robust ν_s^2/G_s factorization over many A_5-invariant metrics (GC32). **This is the terminal version of ZS-F34; the residual computations are deferred to ZS-F35.** This work used Anthropic's Claude; the author assumes responsibility, including for errors caught only on later review.

Appendix A. Numerical Regressions (v1.1)

All from zs_f34_verify_v1_1.py, mpmath 50-digit, locked inputs and standard constants only. (Tags **AC** are this suite; tags **GC** cited in §2.3/§4.3/§6.3/§7 are the gate-decomposition suite zs_f34_verify_v1_8.py, 32/32 — arithmetic, counterexample, and model-instantiation checks.)

Tag	Quantity	Value
AC4	$z^* = i^{z^*}$	0.4382829367 + 0.3605924719 i
AC5	$\|\lambda^*\|$ ($\|\lambda^*\|^2 = 0.794796$)	0.8915135658
AC6	**κ_λ** $= -\ln\|\lambda^*\|$ (loxodromic)	0.1148346250
AC7	$\omega = \arg \lambda^*$	2.2592495539
AC9	Koenigs $\tau = (\omega + i\,\kappa_\lambda)/2\pi$	0.35957 + 0.018276 i
AC10	theta-line $c_1 = \int\omega_2 = 1$ (metric-independent)	1
AC11	**F34.M1**: $V_\Sigma(c_\Sigma=2) = V_\Sigma(1)/4$ (area not fixed)	0.346487 = 1.385949/4
AC12	$V_\Sigma\|_{c_\Sigma=1} = 1/(2\pi\,\kappa_\lambda)$	1.385949
AC13	anti-numerology vs 2 ln 2 = 1.386294	rel 0.000249 (distinct)
AC14	**correction**: 2D even seam-odd torsion ln T	ln 4 = 1.386 ($\neq 0$)
AC15	lattice no-go: $8.190 \in \{a \ln 2 + b \ln 3\}$?	None (closed-negative)
AC16	$g_{reg}^2 = 6\mathbf{A}/\mathbf{Q} = 210/4807$ (κ_{reg}^2)	0.04368629
AC17	$C_M^{sp} = 11 \ln 2 + \ln 3$ (Higgs VEV, $\neq g_6$)	8.7232313
AC19	$\Omega_\Lambda = \tau(P_\Lambda) = 83/121$ (continuous trace; 83 > 11)	0.68595
AC21	general $\mathbf{c}_\theta^T Z^{-1} \mathbf{c}_\theta = 83$ (diagonal-equal only)	83
AC22	**superseded** $\chi_-/e_6^2 = 83\, g_{reg}^2\, \kappa_\lambda/(2\pi)$ [v1.1 Koenigs area $2\pi\,\kappa_\lambda$]	0.066270
AC23	$\rho_{\Lambda,Z} / \chi_- = \omega^2/2$	2.5521043

*AC22 is the v1.1 regression value, computed with the retracted Koenigs area $2\pi\,\kappa_\lambda$ ($V_\Sigma = 1.385949$). The v1.3 metric is the principal polarization (Area = 1, $V_\Sigma = 1$), giving $\chi_-/e_6^2 = 83\, g_{reg}^2/(4\pi^2) =$ **0.091847** (gate check GC11); the two differ by the factor $2\pi\,\kappa_\lambda = 0.7215$.*

REPORTED (excluded): e_6 for $\rho_{obs} \approx (3.219\ \text{meV})^2$ (principal-polarization coefficient; the v1.1-table value used $(3.49\ \text{meV})^2$).

Appendix B. Cross-Version Consistency Audit (v1.7)

Upstream object	Used as	v1.1 verdict
ZS-F32 $\rho\_\Lambda,Z = ½\chi-\omega^2$	§7 link	PASS (AC8, AC23)
ZS-F33.2B Koenigs τ, theta-line $c_1 = 1$	§3, §4 (metric-independent normalization)	PASS (AC9, AC10)
ZS-F33.2B Computation ET (2D seam-odd torsion ln 4)	§5 (retracts even-dim No-Go)	PASS (AC14)
ZS-F33.5 / A28 lattice no-go	§5 (closes scale route)	PASS (AC15)
ZS-M6 $g\_\Gamma^2 = \dim\cdot\kappa\_reg^2$	§5 (register coupling)	PASS, matching OPEN (AC16, OG2)
ZS-S4 C_M^sp	§5 (kept distinct)	PASS (AC17)
ZS-A17 Spin–Metric No-Go	§2 (KK not licensed)	PASS (softened wording)
ZS-A24 II_1 corner, $\tau(P\_\Lambda)$ = 83/121	§6 (83 is a trace)	PASS (AC19)
ZS-A29 rank-83 kinematic precursor	§6 (G2 OPEN)	corrected: v1.0 "proves free" RETRACTED
ZS-A30 one-quantum-per-face	§6 (n_i = 1)	DERIVED-CONDITIONAL (carried)
ZS-M43 No-Thermal-Transport	§10 (Γ_Y on Y, not Z)	PASS (context only)

No upstream result is overturned; v1.0's three overclaims (unique metric, A29 "proves," even-dim No-Go) are corrected to match upstream status.

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Version History

v1.8 (June 2026) — TERMINAL (final-clean): Eighth-review pre-publication pass — no new theorems. An eighth review confirmed the v1.7 content is terminal (32/32) and flagged only publication-clean items, all applied here. **(typesetting)** the \;=\; thick-space artifacts are simplified to plain = (they rendered as ;=; in some viewers); the source LaTeX for Z|_{V^G}, N_{\rm eff}, \hat{\mathbf c}_\theta etc. was already correct (the *-for-_ seen in rendering is a markdown-renderer effect, not a source bug). **(F34.SR precision)** N_eff is sharpened to N_eff = dim S_src ≤ dim V^G for an admissible source space S_src ⊆ V^G, with equality when the admissible sources span V^G (in F34's exact-A_5 branch dim V^{A_5} = 1, so the conclusion is unchanged). **(F34.BIV signature caveat)** F34.BIV is Euclidean ($\star^2 = +1$, a real self-dual/anti-self-dual split); the corpus carrier Λ²(ℝ^{1,3}) is Lorentzian ($\star^2$

= −1 on real 2-forms, ±i eigenspaces after complexification), so the identification needs complexification or an explicit signature-changing intertwiner — added as part of the carrier bridge, not a real-orthogonal identity. **(F34.BIV scope)** recorded that F34.BIV selects the *representation type* $3 \oplus 3'$, while the actual baryon projector P_b retains a residual $\mathbb{CP}^3$ multiplicity-line freedom; fixing P_b is a dynamical (parent-Hessian) question deferred to ZS-F35. **(Outlook sharpened)** §10 now splits the follow-on work into **ZS-F35** (dimensionless/lattice: Multiplicity-Line No-Go, Baryon-Embedding Independence P_s P_b = 0 so I_s is independent of the multiplicity line, physical singlet stiffness Z_s^{phys}, primitive singlet lift, optional approximate-symmetry $O(\varepsilon^2)$ stability) and **ZS-F36** (the sole dimensionful e_6 from a pre-registered UV completion). **(script)** zs_f34_verify_v1_8.py docstring updated to v1.8/GATE-DECOMPOSITION-v5 and the OG3 note changed from "CLOSURE PROGRAMME"/"U_s = V_s" to "REDUCTION"/"(U_s, V_s) = 1"; the 32/32 verification is unchanged. F34 remains terminal.

v1.7 (June 2026) — TERMINAL: Seventh-review fix + two general theorems. A seventh review found **one real formula error** and recommended terminating F34; a parallel deep exploration showed the value is raised most by stating two general theorems. **(error fixed)** the single-mode susceptibility is $\chi_- = (Z\_match\ g\_reg^2/4\pi^2\ V\_\Sigma)\ e_6^2\ (\nu\_s^2/G\_s)$ with $\boldsymbol{\nu\_s^2 = v\_s^T v\_s}$ (lattice norm) and $\boldsymbol{G\_s = s^T G\ s}$ (Hessian norm), since $q\_s = v\_s^T G^{-1} v\_s = \nu\_s^2/G\_s$ already — v1.6 wrongly set $\nu\_s^2 = v\_s^T G^{-1} v\_s$ and divided by G_s again (double-count); GC23 rebuilt on an A_5-invariant SPD metric, GC32 verifies the factorization over many such metrics. **(F34.BIV — new general theorem)** Λ^2(standard 4-dim A_5-module) = $\boldsymbol{3 \oplus 3'}$ (character, GC27); it is the 6-dim S_5-irreducible (GC28); the two triplets are the Hodge self-dual/anti-self-dual sectors, and even permutations commute with the Hodge star (preserve them) while orientation-reversing odd permutations anticommute (swap $3 \leftrightarrow 3'$) (GC29) — this **geometrizes** v1.6's outer automorphism. Cross-checked by the $T_1 \oplus T_2$ decomposition of the 6-dim crystallographic icosahedral representation. Math PROVEN; carrier identification DERIVED-CONDITIONAL. **(F34.SR — new general theorem)** for G-equivariant Z > 0 and source $c \in V^G$, $\chi = (1/4\pi^2)\ c^\dagger (Z|\_{V^G})^{-1} c$, so **N_eff = dim V^G**, not the full flux rank (GC30, GC31); corollaries dim V^G = 0/1/m. Valid for any finite-symmetry multi-four-form theory; the "factor 83" was never the right multiplicity for an invariant source. **(F34.OUT split)** v1.6's single HYPOTHESIS-strong tag is separated into **F34.OUT-Math** (σ-stable selection, DERIVED-CONDITIONAL) and **G-Outer-Physical** (σ = physical Z-Spin symmetry, OPEN); HYPOTHESIS-strong added to the legend. **(I_s)** the single-mode unknowns bundle into the basis-independent **Singlet Response Invariant** $I\_s = Z\_match\ v\_s^T G^{-1} v\_s$, $\chi_- = (g\_reg^2\ e_6^2/4\pi^2\ V\_\Sigma)\ I\_s$. **(branch ratio)** $\chi\_83/\chi\_singlet = 83/q\_s$ (= 83 at q_s = 1), a conditional discriminator. **(wording)** "single-mode closure" → "single-mode reduction"; "the entire dark-energy coefficient reduces to five objects" → "the 83-dimensional flux/source realization reduces to five 1-dim objects {P_s, G_s, ν_s, (U_s, V_s) = 1, Q_s}; the independent G-Metric (V_Σ) and G-Charge (Z_match, e_6) gates remain"; the singlet-holonomy gate sharpened from U_s = V_s to **U_s = V_s = 1**. **(scope)** the residual computations (P_b, I_s, e_6) are deferred to **ZS-F35**; no new closure theorem is added. Added zs_f34_verify_v1_8.py (32/32). Honest framing: the contribution is a **symmetry-reduction theory of multi-top-form vacuum susceptibility** — a *reduction*, not a closure. **F34 terminates here.**

v1.6 (June 2026): Sixth-review correction: outer-automorphism selection + single-mode reduction (v1.6 called this "single-mode closure"; renamed "reduction" in v1.7). A sixth review found three places where v1.5 over-reached, all corrected, plus two new results. **(F34.SEL — selection not unique)** v1.5

said singlet-selection "favours $3 \oplus 3'$." The singlet-free rank-6 A_5-modules are **$\{2\cdot3, 3 \oplus 3', 2\cdot3'\}$** (DE singlet counts 0,1,1,1 for $\{1\oplus5, 2\cdot3, 3\oplus3', 2\cdot3'\}$; GC17, GC18), so F34.SEL only **excludes $1 \oplus 5$** and additionally requires P_b, P_c to be mutually orthogonal **A_5-equivariant** projectors (else the singlet has no representation-theoretic location and one is already in path c). **(F34.OUT — new, outer-automorphism selection)** $\sigma \in \mathrm{Out}(A_5) \cong \mathbb{Z}_2$ (odd permutation) swaps the two 5-cycle classes, hence **σ: $3 \leftrightarrow 3'$, fixing 1, 4, 5** (GC19a); among the singlet-free candidates $2\cdot3 \leftrightarrow 2\cdot3'$ while only **$3 \oplus 3'$ is σ-stable** (GC19b), so A_5 + σ + no-baryon-singlet **uniquely** select baryon = $3 \oplus 3'$. HYPOTHESIS-strong, pending identification of σ with a physical symmetry (seam orientation / J_Z parity / register automorphism / CPT exchange). **(F34.A5-2 — single-mode made precise)** the single-mode $\chi_- = e_6^2/(4\pi^2 Z\_s)$ holds only when DE keeps the singlet **and** $[K, R(g)] = 0$ (singlet decouples, $P\_s K (I-P\_s) = 0$, GC21); if A_5 is broken a singlet source mixes via the **Schur complement** $(Z^{-1})\_ss = 1/(Z\_ss - B D^{-1} B^\dagger)$ (GC22). The singlet **direction** is fixed but its **normalization ν_s is not**, so the susceptibility is $\chi_- = (Z\_match\ g\_reg^2/4\pi^2 V\_\Sigma)\ e_6^2\ (\nu\_s^2/G\_s)$ with $\nu\_s^2 = v\_s^T G^{-1} v\_s$ *(this definition double-counts G_s and is corrected in v1.7: $\nu\_s^2 = v\_s^T v\_s$, $G\_s = s^T G\ s$)* — the 121/F-A24.9 problem persists in the single-mode branch as the one number ν_s (GC23). **(single-mode closure, new)** path (a) reduces the 83-dim seven-gate problem to **five 1-dim objects** $\{P\_s, G\_s, \nu\_s, U\_s = V\_s, Q\_s\}$; full 83-centrality and the factor 83 are needed only in path (c) (GC24, OG3). **(wording)** "generically single-mode" → "single-mode branch (one of three)"; "Z_match in the denominator" sharpened to "denominator of Z, numerator of χ_-." **(LaTeX)** set-brace escaping in $\Sigma\_\{\rho\in\{3,3',4,5\}\}$, thin-space and bold-in-math artifacts cleaned. Added zs_f34_gates_v5_verify.py (26/26). Honest framing: the contribution is the **outer-automorphism selection** of the baryon module and the **single-mode closure reduction** — neither a forced closure nor a No-Go.

v1.5 (June 2026): Fifth-review correction: A_5 representation theory + Z_match; source paths computed (not a No-Go) (the "favours $3 \oplus 3'$" wording and the single-mode conditions later refined in v1.6). **(F34.M4(c) — Schur corrected)** an A_5-invariant $K = \oplus\_\rho (I\_\{d\_\rho\} \otimes K\_\rho)$ has **multiplicity matrices** $K_3, K_{3'} \in M_4$, $K_4 \in M_6$, $K_5 \in M_7$; commutant **dim 117**, up to **21** eigenvalues (GC7, GC8) — correcting v1.4's "block-scalar, ≥4 eigenvalues." **(F34.A5-1 — 83 conditional)** $83 = 4\cdot3 \oplus 4\cdot3' \oplus 6\cdot4 \oplus 7\cdot5$ holds only for the illustrative baryon $6 = 1\oplus5$; End(V_{11}), F_TI character-computed (GC2, GC3). **(F34.SEL, F34.A5-2 — source paths)** a nonzero A_5-invariant θ-coupling exists iff DE keeps the lone singlet; trichotomy (a) single-mode / (b) spurion / (c) factor-83 under alignment; 0.091847 lives in path (c). **(F34.6 — Z_match corrected)** $Z = (V\_\Sigma/Z\_match\ g\_reg^2)\ G$, Z_match in the denominator (v1.4 had it inverted); $c\_\theta = e_6\ \hat{c}\_\theta$. **(F34.M4(b))** "up to 121" removed (full index OPEN). Added zs_f34_gates_v4_verify.py (16/16).

v1.4 (June 2026): Fourth-review correction: kinetic centrality OPEN + carrier minimality (the A_5 Schur claim and the 83-decomposition later corrected in v1.5). **(F34.M4(c) — centrality OPEN)** v1.3 wrongly claimed kinetic centrality follows from the uniform face trace. It does not: with $q\_a = |a\rangle\langle a|$, $\tau(q\_a K q\_b) = (1/121)K\_\{aa\}\delta\_\{ab\}$, so orthogonality gives only **diagonality**; the counterexample $K = \mathrm{diag}(1, \ldots, 83)$ has equal face traces but unequal $Z\_\{aa\}$. *Note: v1.4 then over-simplified the A_5 structure to "block-scalar, ≥4 eigenvalues" and treated $83 = 4\cdot3\oplus4\cdot3'\oplus6\cdot4\oplus7\cdot5$ as fixed; v1.5 corrects both (commutant 117; baryon-conditional).* **(F34.1 — minimal carrier)** form-degree counting gives only $D\_parent \geq 6$, so $M_4 \times \Sigma_2$ ($4 + \dim \mathbf{Z} = 6$) is the **minimal** carrier, unique only under a Minimal-Carrier /

Corpus-Completeness principle; the "not a 6D spacetime" wording is sharpened to "a six-dimensional pseudo-Riemannian total space with four noncompact spacetime dimensions and a compact 2D internal fibre." **(F34.M4(b) — 121 = integral lattice embedding)** recast from "basis ambiguity": a genuine basis change leaves the quadratic form invariant (computed), so the factor 121 is an integral charge-lattice embedding choice, and F-A24.9 is sharpened to a *trace-preserving integral charge-lattice intertwiner*. **(c_θ vs Q_source)** separated. **(F34.M3 — wording)** "closure" → "candidate normalization"; the three levels separated; $c_1 = [iF/2\pi]$ convention stated. **(arithmetic)** ρ_Λ,Z corrected 0.234426 → **0.234402**. **(LaTeX/docs)** parent-action spacing fixed; brace artifacts removed; References A24/A29/A30 synced. Added zs_f34_gates_v3_verify.py (11/11).

v1.3 (June 2026): Third-review correction: principal polarization + sub-gate decomposition (centrality claim and carrier uniqueness later corrected in v1.4). **(F34.M2 RETRACTED)** the v1.2 Koenigs argument is withdrawn as a non sequitur. **(F34.M3, new)** the metric scale is fixed by the **principal polarization** of F33's degree-1 theta line Θ_Z (Area = c_1 = 1 ⇒ V_Σ = 1); coefficient **0.066270 → 0.091847**. **(F34.M4, new)** G-Flux decomposed into sub-gates {monodromy, 121-normalization, kinetic centrality, equal charge, Path A}; the v1.2 closure claims withdrawn. **(F34.1)** Y6 given a degree-counting derivation. **(F34.4-NG)** "non-unitary" corrected to "seam-parity-restricted determinant torsion." Added zs_f34_gates_v2_verify.py (10/10). *Note: v1.3 still wrongly marked kinetic centrality "established" and the carrier "forced"; both corrected in v1.4.*

v1.2 (June 2026): Gate-closure update (now partly superseded by v1.3). Claimed G-Metric CLOSED via a Koenigs-canonical metric (F34.M2 — **retracted in v1.3**) and G-Flux REDUCED to ZS-A24/A30, with $\chi_- = 0.066270 \cdot$ Z_match $\cdot\, e_6^2$ as DERIVED-CONDITIONAL. The third review found the Koenigs argument a non sequitur and a 121-fold normalization ambiguity in the flux coefficient; see the v1.3 entry. Added zs_f34_gates_verify.py (10/10).

v1.1 (June 2026): Major revision integrating an external peer review (MAJOR REVISION verdict). Substantive changes: **(notation)** separated the overloaded symbol κ into κ_λ = −ln|λ*| (loxodromic decrement) and κ_reg² = **A/Q** (register), with g_reg² = 6κ_reg²; **(F34.M1, new NO-GO)** proved that λ*-invariance fixes the torus complex structure but not the Kähler area (c_Σ free), so V_Σ = 1.385949 is COMPUTED-UNDER-NORMALIZATION (c_Σ = 1), not derived, and the corpus theta-line c_1 = 1 is a metric-independent cohomology condition; **(F34.2, new)** added the parent-action wrapped-brane reduction giving Z_{ij}, e_mem = e_6^(6)∫ω_2, T_mem^(4) = T_4-brane Area(Σ_2), the general susceptibility χ_- = (1/4π²) **c**_θ^T Z^{-1} **c**_θ, and the anomaly gates (G-A2); **(F34.4)** demoted the coupling to the register value g_reg² = 6**A/Q** pending a register→EFT matching Z_match (OPEN); **(F34.4-NG, RETRACTED)** withdrew the even-dimensional Ray–Singer No-Go as a category error; **(F34.5, demoted to OPEN-GATE)** corrected the b_2 = 1 → 83 overclaim (83 is the ZS-A24 continuous trace τ(P_Λ), not 83 fluxes), gave the two Realization paths, and withdrew the "BP gap solved" claim; **(F34.6)** recast χ_- = 0.066270 e_6^2 as a BENCHMARK-COROLLARY; **(F34.8)** recast the residual as a finite UV matching problem; **(structure)** moved the thermal history to the Outlook; softened "KK forbidden" to "KK not licensed". Verification reframed: 23/23 checks PASS, explicitly **not** a certificate of physical closure (zs_f34_verify_v1_1.py).

v1.0 (June 2026): Initial public release (superseded). Established the F33.8D ansatz and computed the Charge-Unit reduction, but overclaimed in three places (unique cylinder metric; rank-83 as 83 independent fluxes; register coupling as the continuum coupling) and contained a Ray–Singer dimension category error — all corrected in v1.1.

(**A**, **Q**, dim **Z**) = (35/437, 11, 2) **LOCKED**.